PHYS7045 Quantum Technologies

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1 Assignment 1

1.1 Question 1

1.1.1 Part A

Define the components of the Bloch vector by

$$r_x = \sin\theta\cos\phi, \qquad r_y = \sin\theta\sin\phi, \qquad r_z = \cos\theta$$

For the arbitrary qubit state

$$|\psi\rangle = \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) \\ e^{i\phi}\sin\left(\frac{\theta}{2}\right) \end{pmatrix}$$

the corresponding bra is

$$\langle\psi| = \begin{pmatrix} \cos\left(\frac{\theta}{2}\right), & e^{-i\phi}\sin\left(\frac{\theta}{2}\right) \end{pmatrix}$$

Therefore,

$$\rho = |\psi\rangle\langle\psi| = \begin{pmatrix} \cos^2\left(\frac{\theta}{2}\right) & e^{-i\phi}\cos\left(\frac{\theta}{2}\right)\sin\left(\frac{\theta}{2}\right) \\ e^{i\phi}\cos\left(\frac{\theta}{2}\right)\sin\left(\frac{\theta}{2}\right) & \sin^2\left(\frac{\theta}{2}\right) \end{pmatrix}$$

Using the trigonometric identities:

$$\cos^2\left(\frac{\theta}{2}\right) = \frac{1+\cos\theta}{2}, \qquad \sin^2\left(\frac{\theta}{2}\right) = \frac{1-\cos\theta}{2}, \qquad 2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right) = \sin\theta,$$

we obtain

$$\rho = \frac{1}{2}\begin{pmatrix} 1+\cos\theta & e^{-i\phi}\sin\theta \\ e^{i\phi}\sin\theta & 1-\cos\theta \end{pmatrix}$$

Since

$$e^{\pm i\phi} = \cos\phi \pm i\sin\phi$$

this becomes

$$\rho = \frac{1}{2}\begin{pmatrix} 1+\cos\theta & \sin\theta\cos\phi - i\sin\theta\sin\phi \\ \sin\theta\cos\phi + i\sin\theta\sin\phi & 1-\cos\theta \end{pmatrix}$$

Using the definitions of r_x, r_y, and r_z,

$$\rho = \frac{1}{2}\begin{pmatrix} 1+r_z & r_x - ir_y \\ r_x + ir_y & 1-r_z \end{pmatrix}$$

Now,

$$r_x\sigma_x + r_y\sigma_y + r_z\sigma_z = \begin{pmatrix} r_z & r_x - ir_y \\ r_x + ir_y & -r_z \end{pmatrix}$$

so

$$I_2 + r_x\sigma_x + r_y\sigma_y + r_z\sigma_z = \begin{pmatrix} 1+r_z & r_x - ir_y \\ r_x + ir_y & 1-r_z \end{pmatrix}$$

Hence,

$$\boxed{\rho = \frac{1}{2}\left(I_2 + \mathbf{r}\cdot\boldsymbol{\sigma}\right)}$$

where

$$\mathbf{r}\cdot\boldsymbol{\sigma} = r_x\sigma_x + r_y\sigma_y + r_z\sigma_z$$

1.1.2 Part B

We wish to show that

$$e^{i\theta\,\mathbf{r}\cdot\boldsymbol{\sigma}} = I_2\cos\theta + i(\mathbf{r}\cdot\boldsymbol{\sigma})\sin\theta$$

First, recall that

$$\sigma_x^2 = \sigma_y^2 = \sigma_z^2 = I_2$$

Since

$$\mathbf{r}\cdot\boldsymbol{\sigma} = r_x\sigma_x + r_y\sigma_y + r_z\sigma_z$$

we have

$$(\mathbf{r}\cdot\boldsymbol{\sigma})^2 = (r_x\sigma_x + r_y\sigma_y + r_z\sigma_z)^2$$

Expanding,

$$\begin{aligned}(\mathbf{r}\cdot\boldsymbol{\sigma})^2 = r_x^2\sigma_x^2 + r_y^2\sigma_y^2 + r_z^2\sigma_z^2 \\ + r_x r_y(\sigma_x\sigma_y + \sigma_y\sigma_x) \\ + r_y r_z(\sigma_y\sigma_z + \sigma_z\sigma_y) \\ + r_z r_x(\sigma_z\sigma_x + \sigma_x\sigma_z)\end{aligned}$$

The Pauli matrices anticommute (See Part C), so

$$\sigma_x\sigma_y + \sigma_y\sigma_x = 0 \qquad \sigma_y\sigma_z + \sigma_z\sigma_y = 0 \qquad \sigma_z\sigma_x + \sigma_x\sigma_z = 0$$

Therefore,

$$(\mathbf{r}\cdot\boldsymbol{\sigma})^2 = (r_x^2 + r_y^2 + r_z^2)I_2$$

Hence, for a Bloch vector corresponding to a pure qubit,

$$r_x^2 + r_y^2 + r_z^2 = 1$$

and hence

$$(\mathbf{r}\cdot\boldsymbol{\sigma})^2 = I_2$$

It follows that

$$(\mathbf{r}\cdot\boldsymbol{\sigma})^{2n} = I_2, \qquad (\mathbf{r}\cdot\boldsymbol{\sigma})^{2n+1} = \mathbf{r}\cdot\boldsymbol{\sigma}$$

Using the Taylor expansion of the matrix exponential,

$$e^A = I_2 + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \frac{A^4}{4!} + \cdots$$

with

$$A = i\theta\,\mathbf{r}\cdot\boldsymbol{\sigma},$$

gives

$$\begin{aligned}e^{i\theta\,\mathbf{r}\cdot\boldsymbol{\sigma}} = I_2 + i\theta(\mathbf{r}\cdot\boldsymbol{\sigma}) - \frac{\theta^2}{2!}I_2 - \frac{i\theta^3}{3!}(\mathbf{r}\cdot\boldsymbol{\sigma}) \\ + \frac{\theta^4}{4!}I_2 + \frac{i\theta^5}{5!}(\mathbf{r}\cdot\boldsymbol{\sigma}) - \cdots\end{aligned}$$

Grouping the even and odd powers,

$$\begin{aligned}e^{i\theta\,\mathbf{r}\cdot\boldsymbol{\sigma}} = I_2\left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots\right) \\ + i(\mathbf{r}\cdot\boldsymbol{\sigma})\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots\right)\end{aligned}$$

Recognising the Taylor expansions of cosine and sine,

$$\cos\theta = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots,$$

and

$$\sin\theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots,$$

we obtain

$$\boxed{e^{i\theta\,\mathbf{r}\cdot\boldsymbol{\sigma}} = I_2\cos\theta + i(\mathbf{r}\cdot\boldsymbol{\sigma})\sin\theta}$$

1.1.3 Part C

The Pauli matrices are

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

First,

$$\sigma_x\sigma_y = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = i\sigma_z.$$

Also,

$$\sigma_y\sigma_x = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix} = -i\sigma_z.$$

Hence,

$$\boxed{\sigma_x\sigma_y = -\sigma_y\sigma_x = i\sigma_z}.$$

Next,

$$\sigma_y\sigma_z = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} = i\sigma_x.$$

Also,

$$\sigma_z\sigma_y = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix} = -i\sigma_x.$$

Hence,

$$\boxed{\sigma_y\sigma_z = -\sigma_z\sigma_y = i\sigma_x}.$$

Finally,

$$\sigma_z\sigma_x = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = i\sigma_y.$$

Also,

$$\sigma_x\sigma_z = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = -i\sigma_y.$$

Therefore,

$$\boxed{\sigma_z\sigma_x = -\sigma_x\sigma_z = i\sigma_y}.$$

Part D

For a rotation by an angle α about the axis $m \in \{x, y, z\}$, we define

$$R_m(\alpha) = e^{-i\frac{\alpha}{2}\sigma_m}.$$

Using the result from part (b),

$$e^{i\theta\sigma_m} = I_2\cos\theta + i\sigma_m\sin\theta,$$

we obtain

$$R_m(\alpha) = I_2\cos\left(\frac{\alpha}{2}\right) - i\sigma_m\sin\left(\frac{\alpha}{2}\right).$$

The density operator transforms under this unitary rotation as

$$\rho' = R_m(\alpha)\rho R_m^\dagger(\alpha).$$

From part (a),

$$\rho = \frac{1}{2}\left(I_2 + \mathbf{r}\cdot\boldsymbol{\sigma}\right),$$

so

$$\rho' = \frac{1}{2}R_m\left(I_2 + \mathbf{r}\cdot\boldsymbol{\sigma}\right)R_m^\dagger.$$

Since

$$R_m R_m^\dagger = I_2,$$

this becomes

$$\rho' = \frac{1}{2}\left[I_2 + R_m(\mathbf{r}\cdot\boldsymbol{\sigma})R_m^\dagger\right].$$

Let

$$c = \cos\left(\frac{\alpha}{2}\right), \qquad s = \sin\left(\frac{\alpha}{2}\right).$$

Then

$$R_m = cI_2 - is\sigma_m, \qquad R_m^\dagger = cI_2 + is\sigma_m.$$

Hence,

$$\begin{aligned} R_m(\mathbf{r}\cdot\boldsymbol{\sigma})R_m^\dagger &= (cI_2 - is\sigma_m)(\mathbf{r}\cdot\boldsymbol{\sigma})(cI_2 + is\sigma_m) \\ &= c^2(\mathbf{r}\cdot\boldsymbol{\sigma}) + ics\left[(\mathbf{r}\cdot\boldsymbol{\sigma})\sigma_m - \sigma_m(\mathbf{r}\cdot\boldsymbol{\sigma})\right] \\ &\quad + s^2\sigma_m(\mathbf{r}\cdot\boldsymbol{\sigma})\sigma_m. \end{aligned}$$

We now consider each rotation axis separately.
For $m = x$,

$$R_x = e^{-i\frac{\alpha}{2}\sigma_x}.$$

Using the Pauli matrix relations from part (c),

$$R_x\sigma_x R_x^\dagger = \sigma_x,$$

$$R_x\sigma_y R_x^\dagger = \sigma_y\cos\alpha + \sigma_z\sin\alpha,$$

and

$$R_x\sigma_z R_x^\dagger = -\sigma_y\sin\alpha + \sigma_z\cos\alpha.$$

Therefore,

$$\begin{aligned} R_x(\mathbf{r}\cdot\boldsymbol{\sigma})R_x^\dagger &= r_x\sigma_x + r_y(\sigma_y\cos\alpha + \sigma_z\sin\alpha) \\ &\quad + r_z(-\sigma_y\sin\alpha + \sigma_z\cos\alpha) \\ &= r_x\sigma_x + (r_y\cos\alpha - r_z\sin\alpha)\sigma_y \\ &\quad + (r_y\sin\alpha + r_z\cos\alpha)\sigma_z. \end{aligned}$$

Thus,

$$\rho' = \frac{1}{2}\left(I_2 + r'_x\sigma_x + r'_y\sigma_y + r'_z\sigma_z\right),$$

where

$$r'_x = r_x, \qquad r'_y = r_y\cos\alpha - r_z\sin\alpha, \qquad r'_z = r_y\sin\alpha + r_z\cos\alpha.$$

Hence,

$$\boxed{\begin{pmatrix} r'_x \\ r'_y \\ r'_z \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\alpha & -\sin\alpha \\ 0 & \sin\alpha & \cos\alpha \end{pmatrix}\begin{pmatrix} r_x \\ r_y \\ r_z \end{pmatrix}.}$$

Rotation about the y-axis.
For $m = y$,

$$R_y = e^{-i\frac{\alpha}{2}\sigma_y}.$$

Using the Pauli matrix relations,

$$R_y \sigma_x R_y^\dagger = \sigma_x \cos\alpha - \sigma_z \sin\alpha,$$

$$R_y \sigma_y R_y^\dagger = \sigma_y,$$

and

$$R_y \sigma_z R_y^\dagger = \sigma_x \sin\alpha + \sigma_z \cos\alpha.$$

Therefore,

$$\begin{aligned} R_y(\mathbf{r}\cdot\boldsymbol{\sigma})R_y^\dagger &= r_x(\sigma_x\cos\alpha - \sigma_z\sin\alpha) + r_y\sigma_y \\ &\quad + r_z(\sigma_x\sin\alpha + \sigma_z\cos\alpha) \\ &= (r_x\cos\alpha + r_z\sin\alpha)\sigma_x + r_y\sigma_y \\ &\quad + (-r_x\sin\alpha + r_z\cos\alpha)\sigma_z. \end{aligned}$$

Thus,

$$r'_x = r_x\cos\alpha + r_z\sin\alpha, \qquad r'_y = r_y, \qquad r'_z = -r_x\sin\alpha + r_z\cos\alpha.$$

Hence,

$$\boxed{\begin{pmatrix} r'_x \\ r'_y \\ r'_z \end{pmatrix} = \begin{pmatrix} \cos\alpha & 0 & \sin\alpha \\ 0 & 1 & 0 \\ -\sin\alpha & 0 & \cos\alpha \end{pmatrix} \begin{pmatrix} r_x \\ r_y \\ r_z \end{pmatrix}}.$$

Rotation about the z-axis.

For $m = z$,

$$R_z = e^{-i\frac{\alpha}{2}\sigma_z}.$$

Using the Pauli matrix relations,

$$R_z \sigma_x R_z^\dagger = \sigma_x\cos\alpha + \sigma_y\sin\alpha,$$

$$R_z \sigma_y R_z^\dagger = -\sigma_x\sin\alpha + \sigma_y\cos\alpha,$$

and

$$R_z \sigma_z R_z^\dagger = \sigma_z.$$

Therefore,

$$\begin{aligned} R_z(\mathbf{r}\cdot\boldsymbol{\sigma})R_z^\dagger &= r_x(\sigma_x\cos\alpha + \sigma_y\sin\alpha) \\ &\quad + r_y(-\sigma_x\sin\alpha + \sigma_y\cos\alpha) + r_z\sigma_z \\ &= (r_x\cos\alpha - r_y\sin\alpha)\sigma_x \\ &\quad + (r_x\sin\alpha + r_y\cos\alpha)\sigma_y + r_z\sigma_z. \end{aligned}$$

Thus,

$$r'_x = r_x \cos\alpha - r_y \sin\alpha, \qquad r'_y = r_x \sin\alpha + r_y \cos\alpha, \qquad r'_z = r_z.$$

Hence,

$$\boxed{\begin{pmatrix} r'_x \\ r'_y \\ r'_z \end{pmatrix} = \begin{pmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} r_x \\ r_y \\ r_z \end{pmatrix}}.$$

Therefore,

$$\boxed{R_m(\alpha) = e^{-i\frac{\alpha}{2}\sigma_m}}$$

rotates the Bloch vector by an angle α about the axis m.

1.1.4 Part E

We wish to show that a general rotation can be written in the form

$$R_{\mathbf{n}}(\phi) = e^{i\frac{\phi}{2}(\mathbf{n}\cdot\boldsymbol{\sigma})}$$

where

$$\mathbf{n} = (n_x, n_y, n_z), \qquad n_x^2 + n_y^2 + n_z^2 = 1$$

A general rotation can be represented using Euler angles as

$$R = R_z(\alpha)R_y(\beta)R_z(\gamma).$$

Using the Euler identity from part (b),

$$e^{i\theta\sigma_m} = I_2\cos\theta + i\sigma_m\sin\theta$$

we may write

$$R_z(\alpha) = I_2\cos\left(\frac{\alpha}{2}\right) + i\sigma_z\sin\left(\frac{\alpha}{2}\right)$$

$$R_y(\beta) = I_2\cos\left(\frac{\beta}{2}\right) + i\sigma_y\sin\left(\frac{\beta}{2}\right)$$

and

$$R_z(\gamma) = I_2\cos\left(\frac{\gamma}{2}\right) + i\sigma_z\sin\left(\frac{\gamma}{2}\right)$$

Therefore,

$$\begin{aligned} R = & \left[I_2\cos\left(\frac{\alpha}{2}\right) + i\sigma_z\sin\left(\frac{\alpha}{2}\right)\right] \\ & \times \left[I_2\cos\left(\frac{\beta}{2}\right) + i\sigma_y\sin\left(\frac{\beta}{2}\right)\right] \\ & \times \left[I_2\cos\left(\frac{\gamma}{2}\right) + i\sigma_z\sin\left(\frac{\gamma}{2}\right)\right] \end{aligned}$$

Using the Pauli matrix products from part (c),

$$\sigma_z\sigma_y = -i\sigma_x, \qquad \sigma_y\sigma_z = i\sigma_x, \qquad \sigma_z^2 = I_2,$$

and collecting the coefficients of $I_2, \sigma_x, \sigma_y, \sigma_z$, we obtain

$$R = aI_2 + i\left(b_x\sigma_x + b_y\sigma_y + b_z\sigma_z\right)$$

where

$$a = \cos\left(\frac{\beta}{2}\right)\cos\left(\frac{\alpha+\gamma}{2}\right)$$

$$b_x = -\sin\left(\frac{\beta}{2}\right)\sin\left(\frac{\alpha-\gamma}{2}\right)$$
$$b_y = \sin\left(\frac{\beta}{2}\right)\cos\left(\frac{\alpha-\gamma}{2}\right)$$

and

$$b_z = \cos\left(\frac{\beta}{2}\right)\sin\left(\frac{\alpha+\gamma}{2}\right)$$

These coefficients satisfy

$$\begin{aligned} a^2 + b_x^2 + b_y^2 + b_z^2 = &\cos^2\left(\frac{\beta}{2}\right)\left[\cos^2\left(\frac{\alpha+\gamma}{2}\right) + \sin^2\left(\frac{\alpha+\gamma}{2}\right)\right] \\ &+ \sin^2\left(\frac{\beta}{2}\right)\left[\sin^2\left(\frac{\alpha-\gamma}{2}\right) + \cos^2\left(\frac{\alpha-\gamma}{2}\right)\right] \\ = &\cos^2\left(\frac{\beta}{2}\right) + \sin^2\left(\frac{\beta}{2}\right) \\ = &1 \end{aligned}$$

We can therefore define an angle ϕ such that

$$a = \cos\left(\frac{\phi}{2}\right)$$

and write

$$b_x = n_x \sin\left(\frac{\phi}{2}\right), \qquad b_y = n_y \sin\left(\frac{\phi}{2}\right), \qquad b_z = n_z \sin\left(\frac{\phi}{2}\right),$$

where

$$n_x^2 + n_y^2 + n_z^2 = 1.$$

Hence,

$$R = I_2 \cos\left(\frac{\phi}{2}\right) + i\left(n_x\sigma_x + n_y\sigma_y + n_z\sigma_z\right)\sin\left(\frac{\phi}{2}\right)$$

Since

$$\mathbf{n}\cdot\boldsymbol{\sigma} = n_x\sigma_x + n_y\sigma_y + n_z\sigma_z$$

we have

$$R = I_2 \cos\left(\frac{\phi}{2}\right) + i(\mathbf{n}\cdot\boldsymbol{\sigma})\sin\left(\frac{\phi}{2}\right)$$

Using the Euler identity from part (b), this is

$$\boxed{R_{\mathbf{n}}(\phi) = e^{i\frac{\phi}{2}(\mathbf{n}\cdot\boldsymbol{\sigma})}}$$

Thus any general qubit rotation can be represented as a single rotation through an angle ϕ about an arbitrary unit axis $\mathbf{n}$.

1.1.5 Part F

Consider two arbitrary states $|\psi\rangle$ and $|\phi\rangle$ and a blank state $|\psi_0\rangle$. If $\hat{Q}$ is a universal cloning operator, then

$$\hat{Q}\left(|\psi\rangle \otimes |\psi_0\rangle\right) = |\psi\rangle \otimes |\psi\rangle,$$

and

$$\hat{Q}\left(|\phi\rangle \otimes |\psi_0\rangle\right) = |\phi\rangle \otimes |\phi\rangle.$$

If we take the inner product of each side of these equations we get:

$$\left(\langle\psi| \otimes \langle\psi_0|\right)\hat{Q}^\dagger\hat{Q}\left(|\phi\rangle \otimes |\psi_0\rangle\right) = \left(\langle\psi| \otimes \langle\psi|\right)\left(|\phi\rangle \otimes |\phi\rangle\right).$$

Since $\hat{Q}$ is unitary,

$$\left(\langle\psi| \otimes \langle\psi_0|\right)\left(|\phi\rangle \otimes |\psi_0\rangle\right) = \left(\langle\psi| \otimes \langle\psi|\right)\left(|\phi\rangle \otimes |\phi\rangle\right).$$

Using the inner product of tensor-product states,

$$\langle\psi|\phi\rangle\langle\psi_0|\psi_0\rangle = \langle\psi|\phi\rangle\langle\psi|\phi\rangle.$$

Since the blank state is normalised,

$$\langle\psi_0|\psi_0\rangle = 1,$$

so

$$\langle\psi|\phi\rangle = \left(\langle\psi|\phi\rangle\right)^2.$$

Let

$$c = \langle\psi|\phi\rangle.$$

Then

$$c = c^2,$$

and hence

$$c(c-1) = 0.$$

Therefore,

$$\langle\psi|\phi\rangle = 0 \qquad \text{or} \qquad \langle\psi|\phi\rangle = 1.$$

Thus the cloning operation can only work when the two possible input states are either orthogonal,

$$\langle\psi|\phi\rangle = 0,$$

or identical,

$$\langle\psi|\phi\rangle = 1.$$

Since arbitrary quantum states need not be orthogonal or identical, no single unitary operator can clone an arbitrary unknown quantum state. Therefore, there is no universal unitary quantum cloning operator.

1.2 Question 2

1.2.1 Part A

For a time-independent Hamiltonian, the time-evolution operator is

$$U(t) = e^{-iHt/\hbar}.$$

Given

$$H = \hbar \left(t_c \sigma_x + \Delta \sigma_z \right),$$

the evolution operator is therefore

$$U(t) = e^{-i(t_c \sigma_x + \Delta \sigma_z)t}.$$

From the previous question, a rotation by an angle α about the axis m is

$$R_m(\alpha) = e^{-i\frac{\alpha}{2}\sigma_m}.$$

To produce a rotation about the x-axis, set

$$\Delta = 0.$$

Then

$$H = \hbar t_c \sigma_x,$$

and

$$U(t) = e^{-it_c t \sigma_x}.$$

Comparing with

$$R_x(\alpha) = e^{-i\frac{\alpha}{2}\sigma_x},$$

we require

$$t_c t = \frac{\alpha}{2}.$$

Hence the rotation angle is

$$\alpha = 2t_c t.$$

For an X_π gate,

$$\alpha = \pi,$$

so

$$t_{X_\pi} = \frac{\pi}{2t_c}.$$

The maximum allowed value of t_c is

$$\frac{t_c}{2\pi} = 5\,\mathrm{GHz},$$

so

$$t_c = 2\pi(5 \times 10^9)\,\mathrm{rad\,s^{-1}}.$$

Therefore,

$$\begin{aligned} t_{X_\pi} &= \frac{\pi}{2\,[2\pi(5 \times 10^9)]} \\ &= 5.0 \times 10^{-11}\,\mathrm{s}. \end{aligned}$$

Thus,

$$\boxed{t_{X_\pi} = 50\,\mathrm{ps}}.$$

To produce a rotation about the z-axis, set

$$t_c = 0.$$

Then

$$H = \hbar\Delta\sigma_z,$$

and

$$U(t) = e^{-i\Delta t\sigma_z}.$$

Comparing with

$$R_z(\alpha) = e^{-i\frac{\alpha}{2}\sigma_z},$$

we require

$$\Delta t = \frac{\alpha}{2},$$

so

$$\alpha = 2\Delta t.$$

For a Z_π gate,

$$\alpha = \pi,$$

and therefore

$$t_{Z_\pi} = \frac{\pi}{2\Delta}.$$

The maximum allowed value of Δ is

$$\frac{\Delta}{2\pi} = 40\,\mathrm{GHz},$$

so

$$\Delta = 2\pi(40 \times 10^9)\,\mathrm{rad\,s^{-1}}.$$

Hence,

$$\begin{aligned} t_{Z_\pi} &= \frac{\pi}{2\,[2\pi(40 \times 10^9)]} \\ &= 6.25 \times 10^{-12}\,\mathrm{s}. \end{aligned}$$

Thus,

$$\boxed{t_{Z_\pi} = 6.25\,\mathrm{ps}}.$$

Therefore, the fastest possible gates are

$$\boxed{X_\pi :\ 50\,\mathrm{ps} \qquad \text{and} \qquad Z_\pi :\ 6.25\,\mathrm{ps}.}$$

Part B

A Y_π rotation can be implemented as a sequence of z- and x-axis rotations:

$$R_y(\pi) = R_z\left(\frac{\pi}{2}\right) R_x(\pi) R_z\left(\frac{3\pi}{2}\right).$$

The first z-rotation maps the x-axis onto the y-axis. The X_π rotation is then performed about this rotated axis, and the final z-rotation maps the axis back to the original coordinate system.

Using the maximum rotation rates from part (a),

$$t_{Z_{\pi/2}} = \frac{1}{2} t_{Z_\pi} = 3.125\,\mathrm{ps},$$

$$t_{X_\pi} = 50\,\mathrm{ps},$$

and

$$t_{Z_{3\pi/2}} = \frac{3}{2} t_{Z_\pi} = 9.375\,\mathrm{ps}.$$

Therefore, the total time is

$$t_{Y_\pi} = 3.125 + 50 + 9.375 = \boxed{62.5\,\mathrm{ps}}.$$

Part C

The Hadamard gate can be implemented using x- and z-axis rotations as

$$H = iR_z\left(\frac{\pi}{2}\right) R_x\left(\frac{\pi}{2}\right) R_z\left(\frac{\pi}{2}\right),$$

where the factor i is a global phase and therefore has no physical effect.

Thus, the Hadamard gate can be implemented by the sequence

$$\boxed{R_z\left(\frac{\pi}{2}\right) \to R_x\left(\frac{\pi}{2}\right) \to R_z\left(\frac{\pi}{2}\right)}.$$

Using the maximum rotation rates from part (a),

$$t_{Z_{\pi/2}} = 3.125\,\text{ps},$$

and

$$t_{X_{\pi/2}} = \frac{1}{2} t_{X_\pi} = 25\,\text{ps}.$$

Therefore, the total time required is

$$t_H = 3.125 + 25 + 3.125 = \boxed{31.25\,\text{ps}}.$$

Part D

For a rotation about the x-axis,

$$\alpha = 2t_c t,$$

while for a rotation about the z-axis,

$$\alpha = 2\Delta t.$$

Since the pulse controller has a time resolution of 1 ns, we choose each elementary rotation to take

$$t = 1\,\text{ns}.$$

Thus, for an X_α rotation,

$$\frac{t_c}{2\pi} = \frac{\alpha}{4\pi t},$$

and for a Z_α rotation,

$$\frac{\Delta}{2\pi} = \frac{\alpha}{4\pi t}.$$

For $t = 1\,\text{ns}$, the required values are therefore

$$X_{\pi/2}: \qquad \frac{t_c}{2\pi} = \frac{\pi/2}{4\pi(1\times10^{-9})} = 0.125\,\text{GHz},$$

$$X_{\pi}: \qquad \frac{t_c}{2\pi} = \frac{\pi}{4\pi(1\times10^{-9})} = 0.250\,\text{GHz},$$

$$Z_{\pi/2}: \qquad \frac{\Delta}{2\pi} = \frac{\pi/2}{4\pi(1\times10^{-9})} = 0.125\,\text{GHz},$$

$$Z_{\pi}: \qquad \frac{\Delta}{2\pi} = \frac{\pi}{4\pi(1\times10^{-9})} = 0.250\,\text{GHz},$$

and

$$Z_{3\pi/2}: \qquad \frac{\Delta}{2\pi} = \frac{3\pi/2}{4\pi(1\times10^{-9})} = 0.375\,\text{GHz}.$$

All of these values lie well within the constraints

$$0 \le \frac{t_c}{2\pi} \le 5\,\text{GHz}, \qquad 0 \le \frac{\Delta}{2\pi} \le 40\,\text{GHz}.$$

Using the decompositions

$$H \sim R_z\left(\frac{\pi}{2}\right) R_x\left(\frac{\pi}{2}\right) R_z\left(\frac{\pi}{2}\right),$$

$$R_y(\pi) = R_z\left(\frac{\pi}{2}\right) R_x(\pi) R_z\left(-\frac{\pi}{2}\right),$$

and

$$R_y\left(\frac{\pi}{2}\right) = R_z\left(\frac{\pi}{2}\right) R_x\left(\frac{\pi}{2}\right) R_z\left(-\frac{\pi}{2}\right),$$

where

$$R_z\left(-\frac{\pi}{2}\right) \sim R_z\left(\frac{3\pi}{2}\right),$$

the required pulse sequence, in chronological order, is

$$\begin{aligned}
H: &\quad Z_{\pi/2},\ X_{\pi/2},\ Z_{\pi/2},\\
X_\pi: &\quad X_\pi,\\
Y_\pi: &\quad Z_{3\pi/2},\ X_\pi,\ Z_{\pi/2},\\
Z_{\pi/2}: &\quad Z_{\pi/2},\\
X_{\pi/2}: &\quad X_{\pi/2},\\
Y_{\pi/2}: &\quad Z_{3\pi/2},\ X_{\pi/2},\ Z_{\pi/2},\\
Z_\pi: &\quad Z_\pi.
\end{aligned}$$

A timing diagram for the control parameters is therefore

Time (ns)	Operation	$t_c/(2\pi)$ (GHz)	$\Delta/(2\pi)$ (GHz)
$0-1$	$Z_{\pi/2}$	0	0.125
$1-2$	$X_{\pi/2}$	0.125	0
$2-3$	$Z_{\pi/2}$	0	0.125
$3-4$	X_π	0.250	0
$4-5$	$Z_{3\pi/2}$	0	0.375
$5-6$	X_π	0.250	0
$6-7$	$Z_{\pi/2}$	0	0.125
$7-8$	$Z_{\pi/2}$	0	0.125
$8-9$	$X_{\pi/2}$	0.125	0
$9-10$	$Z_{3\pi/2}$	0	0.375
$10-11$	$X_{\pi/2}$	0.125	0
$11-12$	$Z_{\pi/2}$	0	0.125
$12-13$	Z_π	0	0.250

Thus, the complete gate sequence requires

$$\boxed{13\,\text{ns}}.$$

During the measurement phase we set

$$\boxed{t_c = 0, \qquad \Delta = 0,}$$

so that the Hamiltonian vanishes and the qubit state does not evolve during the measurement.

Part E

To implement a CZ gate, set

$$t_c = 0, \qquad \Delta = 0,$$

and turn on the interaction C. Then the Hamiltonian of qubit 2 is

$$H_2 = \begin{cases} \hbar\delta_0\sigma_z, & \text{if qubit 1 is } |0\rangle, \\ \hbar\delta_1\sigma_z, & \text{if qubit 1 is } |1\rangle. \end{cases}$$

Since a Hamiltonian

$$H = \hbar\omega\sigma_z$$

produces a z-rotation

$$\theta = 2\omega t,$$

the two conditional rotations are

$$\theta_0 = 2\delta_0 t, \qquad \theta_1 = 2\delta_1 t.$$

For a CZ gate, these rotations must differ by π:

$$2(\delta_1 - \delta_0)t = \pi.$$

Using

$$\frac{\delta_0}{2\pi} = 1\,\text{MHz}, \qquad \frac{\delta_1}{2\pi} = 51\,\text{MHz},$$

gives

$$\delta_1 - \delta_0 = 2\pi(50 \times 10^6)\,\text{rad}\,\text{s}^{-1},$$

and therefore

$$t_C = \frac{\pi}{2(\delta_1 - \delta_0)} = 5\,\text{ns}.$$

Thus,

$$\boxed{t_c = 0, \qquad \Delta = 0, \qquad C = 1 \quad \text{for } 5\,\text{ns}.}$$

During this time,

$$\theta_0 = 0.02\pi, \qquad \theta_1 = 1.02\pi,$$

so that

$$\theta_1 - \theta_0 = \pi.$$

The C-pulse therefore produces the required CZ interaction together with residual single-qubit rotations

$$R_z^{(1)}\left(-\frac{\pi}{2}\right) \qquad \text{and} \qquad R_z^{(2)}(0.02\pi).$$

These are removed by the inverse rotations

$$R_z^{(1)}\left(\frac{\pi}{2}\right) \qquad \text{and} \qquad R_z^{(2)}(-0.02\pi).$$

Since the pulse controller can only change the Hamiltonian parameters in 1 ns time steps, both compensating rotations are implemented during a single 1 ns interval by choosing suitable values of Δ.

For qubit 1,

$$R_z\left(\frac{\pi}{2}\right)$$

requires

$$\frac{\Delta_1}{2\pi} = \frac{\pi/2}{4\pi(1\times 10^{-9})} = \boxed{0.125\,\text{GHz}}.$$

For qubit 2, only positive values of Δ are available, so

$$R_z(-0.02\pi) \sim R_z(1.98\pi),$$

up to a global phase. Therefore,

$$\frac{\Delta_2}{2\pi} = \frac{1.98\pi}{4\pi(1\times 10^{-9})} = \boxed{0.495\,\text{GHz}}.$$

Both compensating rotations can be applied simultaneously. Hence the full CZ implementation is

Time	$t_{c,1}/(2\pi)$	$\Delta_1/(2\pi)$	$t_{c,2}/(2\pi)$	$\Delta_2/(2\pi)$	C
$0-5$ ns	0	0	0	0	1
$5-6$ ns	0	0.125	0	0.495	0

where the frequencies are in GHz.

Therefore, respecting the 1 ns timing resolution, the complete CZ gate takes

$$\boxed{6\,\text{ns}}.$$

1.2.2 Part F

A controlled-X gate can be implemented from a controlled-Z gate by applying Hadamard gates to the target qubit before and after the controlled-Z:

$$\boxed{CX = (I_2 \otimes H)\, CZ\, (I_2 \otimes H)}.$$

This follows from

$$HZH = X.$$

Therefore, the first Hadamard changes the basis of the target qubit so that the controlled-Z acts as a controlled-X, and the second Hadamard changes the target qubit back to the computational basis.

Using the Hadamard implementation from the previous part,

$$H \sim R_z\left(\frac{\pi}{2}\right) R_x\left(\frac{\pi}{2}\right) R_z\left(\frac{\pi}{2}\right),$$

the pulse sequence is

$$H_2 \;\longrightarrow\; CZ \;\longrightarrow\; H_2.$$

Hence a CX gate can be implemented using the same x- and z-axis rotations together with the CZ interaction developed in part (a).

1.2.3 Part G

The circuit is

$$H_1 \;\longrightarrow\; CX_{1,2} \;\longrightarrow\; Y_\pi^{(2)} \;\longrightarrow\; CZ_{1,2} \;\longrightarrow\; \text{measurement}.$$

Using the results from the previous parts,

$$H \sim Z_{\pi/2} X_{\pi/2} Z_{\pi/2},$$

$$CX_{1,2} = H_2\, CZ_{1,2}\, H_2,$$

and

$$Y_\pi = Z_{\pi/2} X_\pi Z_{3\pi/2}.$$

For 1 ns single-qubit pulses,

$$X_{\pi/2}: \qquad \frac{t_c}{2\pi} = 0.125\,\text{GHz},$$

$$X_\pi : \qquad \frac{t_c}{2\pi} = 0.250\,\text{GHz},$$

$$Z_{\pi/2} : \qquad \frac{\Delta}{2\pi} = 0.125\,\text{GHz},$$

and

$$Z_{3\pi/2} : \qquad \frac{\Delta}{2\pi} = 0.375\,\text{GHz}.$$

From part (a), each CZ gate is implemented by turning on C for

$$5\,\text{ns}$$

with

$$t_c = 0, \qquad \Delta = 0,$$

followed by a 1 ns compensation pulse with

$$\frac{\Delta_1}{2\pi} = 0.125\,\text{GHz}, \qquad \frac{\Delta_2}{2\pi} = 0.495\,\text{GHz}.$$

The complete timing diagram is therefore

Time (ns)	Operation	$t_{c,1}/(2\pi)$	$\Delta_1/(2\pi)$	$t_{c,2}/(2\pi)$	$\Delta_2/(2\pi)$	C
$0-1$	$Z^{(1)}_{\pi/2}$	0	0.125	0	0	0
$1-2$	$X^{(1)}_{\pi/2}$	0.125	0	0	0	0
$2-3$	$Z^{(1)}_{\pi/2}$	0	0.125	0	0	0
$3-4$	$Z^{(2)}_{\pi/2}$	0	0	0	0.125	0
$4-5$	$X^{(2)}_{\pi/2}$	0	0	0.125	0	0
$5-6$	$Z^{(2)}_{\pi/2}$	0	0	0	0.125	0
$6-11$	CZ interaction	0	0	0	0	1
$11-12$	CZ compensation	0	0.125	0	0.495	0
$12-13$	$Z^{(2)}_{\pi/2}$	0	0	0	0.125	0
$13-14$	$X^{(2)}_{\pi/2}$	0	0	0.125	0	0
$14-15$	$Z^{(2)}_{\pi/2}$	0	0	0	0.125	0
$15-16$	$Z^{(2)}_{3\pi/2}$	0	0	0	0.375	0
$16-17$	$X^{(2)}_{\pi}$	0	0	0.250	0	0
$17-18$	$Z^{(2)}_{\pi/2}$	0	0	0	0.125	0
$18-23$	CZ interaction	0	0	0	0	1
$23-24$	CZ compensation	0	0.125	0	0.495	0
24+	Measurement	0	0	0	0	0

where all frequencies are in GHz.
Thus the complete circuit is implemented in

$$\boxed{24\,\text{ns}}.$$

During measurement we set

$$\boxed{t_{c,1} = t_{c,2} = 0, \qquad \Delta_1 = \Delta_2 = 0, \qquad C = 0,}$$

so that neither qubit evolves during the measurement.

1.3 Question 3

1.3.1 Part A

The Schmidt decomposition expresses a bipartite pure state in the form

$$|\Psi\rangle = \sum_i \sqrt{\lambda_i}\, |u_i\rangle_A |v_i\rangle_B,$$

where $\{|u_i\rangle_A\}$ and $\{|v_i\rangle_B\}$ are orthonormal sets. Physically, it identifies the natural correlated basis states of the two subsystems. The Schmidt coefficients $\sqrt{\lambda_i}$ quantify the entanglement: the state is separable if only one coefficient is non-zero, and maximally entangled when the non-zero coefficients are equal.

1.3.2 Part B

The Schmidt decomposition is not completely unique. The Schmidt coefficients are unique if we ignore their ordering, but the Schmidt basis states may be multiplied by opposite phase factors without changing the state:

$$|u_i\rangle_A \to e^{i\phi_i}|u_i\rangle_A, \qquad |v_i\rangle_B \to e^{-i\phi_i}|v_i\rangle_B.$$

If we ignore this phase factor and the ability to reorder the coefficients, the Schmidt decomposition is essentially unique.

1.3.3 Part C

Starting from

$$|\Psi\rangle = \frac{1}{\sqrt{2}} \left(|\alpha_1\rangle_A |0\rangle_B + |\alpha_2\rangle_A |1\rangle_B\right),$$

with

$$|\alpha_1\rangle = \cos\theta\, |0\rangle + \sin\theta\, |1\rangle, \qquad |\alpha_2\rangle = \cos\theta\, |0\rangle - \sin\theta\, |1\rangle,$$

we obtain

$$|\Psi\rangle = \frac{1}{\sqrt{2}} \left[(\cos\theta|0\rangle + \sin\theta|1\rangle)|0\rangle + (\cos\theta|0\rangle - \sin\theta|1\rangle)|1\rangle\right].$$

Collecting terms according to subsystem A,

$$|\Psi\rangle = \frac{\cos\theta}{\sqrt{2}}|0\rangle_A \left(|0\rangle_B + |1\rangle_B\right) + \frac{\sin\theta}{\sqrt{2}}|1\rangle_A \left(|0\rangle_B - |1\rangle_B\right).$$

Using

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}},$$

this becomes

$$\boxed{|\Psi\rangle = \cos\theta\,|0\rangle_A|+\rangle_B + \sin\theta\,|1\rangle_A|-\rangle_B}.$$

Since $\{|0\rangle, |1\rangle\}$ and $\{|+\rangle, |-\rangle\}$ are both orthonormal sets, this is the Schmidt decomposition of $|\Psi\rangle$, with $\lambda_1 = \cos^2\theta$ and $\lambda_2 = \sin^2\theta$.

1.3.4 Part D

From part (c), the Schmidt decomposition is

$$|\Psi\rangle = \cos\theta\,|0\rangle_A|+\rangle_B + \sin\theta\,|1\rangle_A|-\rangle_B.$$

The density operator of the full system is

$$\rho_{AB} = |\Psi\rangle\langle\Psi|.$$

Expanding,

$$\begin{aligned}\rho_{AB} = {} & \cos^2\theta\,|0\rangle_A\langle 0| \otimes |+\rangle_B\langle +| \\ & + \cos\theta\sin\theta\,|0\rangle_A\langle 1| \otimes |+\rangle_B\langle -| \\ & + \cos\theta\sin\theta\,|1\rangle_A\langle 0| \otimes |-\rangle_B\langle +| \\ & + \sin^2\theta\,|1\rangle_A\langle 1| \otimes |-\rangle_B\langle -|.\end{aligned}$$

The reduced density matrix for subsystem B is obtained by taking the partial trace over subsystem A:

$$\rho_B = \mathrm{Tr}_A(\rho_{AB}) = \sum_{i=0}^{1} {}_A\langle i|\rho_{AB}|i\rangle_A.$$

Using

$$\langle 0|0\rangle = \langle 1|1\rangle = 1, \qquad \langle 0|1\rangle = \langle 1|0\rangle = 0,$$

the cross terms vanish, giving

$$\boxed{\rho_B = \cos^2\theta\,|+\rangle\langle +| + \sin^2\theta\,|-\rangle\langle -|}.$$

In the computational basis,

$$|+\rangle\langle+| = \frac{1}{2}\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}, \qquad |-\rangle\langle-| = \frac{1}{2}\begin{pmatrix}1 & -1\\ -1 & 1\end{pmatrix},$$

so

$$\rho_B = \frac{1}{2}\begin{pmatrix}\cos^2\theta + \sin^2\theta & \cos^2\theta - \sin^2\theta\\ \cos^2\theta - \sin^2\theta & \cos^2\theta + \sin^2\theta\end{pmatrix}$$

$$\boxed{\rho_B = \frac{1}{2}\begin{pmatrix}1 & \cos 2\theta\\ \cos 2\theta & 1\end{pmatrix}}.$$

1.3.5 Part E

Let p_1 be the probability that Alice successfully identifies $|\alpha_1\rangle$ when her qubit is in the state $|\alpha_1\rangle$, and let p_2 be the corresponding probability for $|\alpha_2\rangle$.

From part (d), Bob's reduced density matrix is

$$\rho_B = \frac{1}{2}\begin{pmatrix} 1 & \cos 2\theta \\ \cos 2\theta & 1 \end{pmatrix}.$$

If Alice successfully identifies $|\alpha_1\rangle$, Bob's qubit is projected onto $|0\rangle$. Similarly, if she successfully identifies $|\alpha_2\rangle$, Bob's qubit is projected onto $|1\rangle$.

Since $|\alpha_1\rangle$ and $|\alpha_2\rangle$ occur with equal probability, the successful outcomes contribute

$$\rho_B^{(\text{succ})} = \frac{p_1}{2}|0\rangle\langle 0| + \frac{p_2}{2}|1\rangle\langle 1|.$$

In matrix form,

$$\rho_B^{(\text{succ})} = \frac{1}{2}\begin{pmatrix} p_1 & 0 \\ 0 & p_2 \end{pmatrix}.$$

By the no-signalling theorem, Alice's local measurement cannot change Bob's overall reduced density matrix. Therefore, the contribution corresponding to Alice's inconclusive outcome must be

$$\rho_B^{(\text{fail})} = \rho_B - \rho_B^{(\text{succ})}.$$

Hence,

$$\rho_B^{(\text{fail})} = \frac{1}{2}\begin{pmatrix} 1-p_1 & \cos 2\theta \\ \cos 2\theta & 1-p_2 \end{pmatrix}.$$

For this to represent a physical state, it must be positive semidefinite. Therefore,

$$\det\left(\rho_B^{(\text{fail})}\right) \geq 0.$$

This gives

$$\frac{1}{4}\left[(1-p_1)(1-p_2) - \cos^2 2\theta\right] \geq 0,$$

and therefore

$$(1-p_1)(1-p_2) \geq \cos^2 2\theta.$$

The total probability of successful unambiguous discrimination is

$$P_{\text{succ}} = \frac{p_1 + p_2}{2}.$$

Define

$$q_1 = 1 - p_1, \qquad q_2 = 1 - p_2.$$

Then

$$q_1 q_2 \geq \cos^2 2\theta.$$

Using the arithmetic–geometric mean inequality,

$$\frac{q_1 + q_2}{2} \geq \sqrt{q_1 q_2} \geq |\cos 2\theta|.$$

Since

$$P_{\text{succ}} = 1 - \frac{q_1 + q_2}{2},$$

we therefore obtain

$$\boxed{P_{\text{succ}} \leq 1 - |\cos 2\theta|}.$$

Finally, since

$$\langle \alpha_1 | \alpha_2 \rangle = \cos 2\theta,$$

the upper bound may also be written as

$$\boxed{P_{\text{succ}} \leq 1 - |\langle \alpha_1 | \alpha_2 \rangle|}.$$

Thus the maximum probability for unambiguous state discrimination is

$$\boxed{P_{\text{USD}}^{\max} = 1 - |\cos 2\theta|}.$$

1.4 Question 4

1.4.1 Part A

Write the unknown qubit as

$$|a\rangle = \alpha|0\rangle + \gamma|1\rangle,$$

where

$$\alpha = \cos\left(\frac{\theta}{2}\right), \qquad \gamma = e^{i\phi}\sin\left(\frac{\theta}{2}\right).$$

The shared entangled state is

$$|AB\rangle = \cos\beta\,|0\rangle_A|1\rangle_B - \sin\beta\,|1\rangle_A|0\rangle_B.$$

Hence

$$|a\rangle|AB\rangle = (\alpha|0\rangle + \gamma|1\rangle)\,(\cos\beta\,|01\rangle - \sin\beta\,|10\rangle)\,.$$

Expanding,

$$\begin{aligned}|a\rangle|AB\rangle = {} & \alpha\cos\beta\,|00\rangle_A|1\rangle_B - \alpha\sin\beta\,|01\rangle_A|0\rangle_B \\ & + \gamma\cos\beta\,|10\rangle_A|1\rangle_B - \gamma\sin\beta\,|11\rangle_A|0\rangle_B.\end{aligned}$$

For the Bell state

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}},$$

we have

$${}_A\langle\Phi^+| = \frac{\langle 00| + \langle 11|}{\sqrt{2}}.$$

Therefore

$$\begin{aligned}{}_A\langle\Phi^+|\,(|a\rangle|AB\rangle) &= \frac{1}{\sqrt{2}}\,(\alpha\cos\beta\,|1\rangle_B - \gamma\sin\beta\,|0\rangle_B) \\ &= \boxed{\frac{1}{\sqrt{2}}\,(-\gamma\sin\beta\,|0\rangle_B + \alpha\cos\beta\,|1\rangle_B)}.\end{aligned}$$

For

$$|\Phi^-\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}},$$

we have

$${}_A\langle\Phi^-| = \frac{\langle 00| - \langle 11|}{\sqrt{2}}.$$

Therefore

$$
\begin{aligned}
{}_A\langle\Phi^-|\left(|a\rangle|AB\rangle\right) &= \frac{1}{\sqrt{2}}\left(\alpha\cos\beta\,|1\rangle_B + \gamma\sin\beta\,|0\rangle_B\right) \\
&= \boxed{\frac{1}{\sqrt{2}}\left(\gamma\sin\beta\,|0\rangle_B + \alpha\cos\beta\,|1\rangle_B\right)}.
\end{aligned}
$$

For

$$
|\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}},
$$

we have

$$
{}_A\langle\Psi^+| = \frac{\langle 01| + \langle 10|}{\sqrt{2}}.
$$

Therefore

$$
\begin{aligned}
{}_A\langle\Psi^+|\left(|a\rangle|AB\rangle\right) &= \frac{1}{\sqrt{2}}\left(-\alpha\sin\beta\,|0\rangle_B + \gamma\cos\beta\,|1\rangle_B\right) \\
&= \boxed{\frac{1}{\sqrt{2}}\left(-\alpha\sin\beta\,|0\rangle_B + \gamma\cos\beta\,|1\rangle_B\right)}.
\end{aligned}
$$

For

$$
|\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}},
$$

we have

$$
{}_A\langle\Psi^-| = \frac{\langle 01| - \langle 10|}{\sqrt{2}}.
$$

Therefore

$$
\begin{aligned}
{}_A\langle\Psi^-|\left(|a\rangle|AB\rangle\right) &= \frac{1}{\sqrt{2}}\left(-\alpha\sin\beta\,|0\rangle_B - \gamma\cos\beta\,|1\rangle_B\right) \\
&= \boxed{-\frac{1}{\sqrt{2}}\left(\alpha\sin\beta\,|0\rangle_B + \gamma\cos\beta\,|1\rangle_B\right)}.
\end{aligned}
$$

1.4.2 Part B

Alice first performs a Bell-basis measurement on her two qubits. Depending on the result, Bob's qubit is projected into one of the four states found in part (a).

Alice then sends Bob the two classical bits specifying which Bell state she measured. Bob uses this information to apply the corresponding Pauli correction to his qubit.

For the outcome $|\Phi^+\rangle$, Bob has

$$|\tilde{b}_{\Phi^+}\rangle = \frac{1}{\sqrt{2}}\left(-\gamma \sin\beta\,|0\rangle + \alpha\cos\beta\,|1\rangle\right).$$

Alice tells Bob that she measured $|\Phi^+\rangle$, so Bob applies the correction ZX. This gives

$$ZX|\tilde{b}_{\Phi^+}\rangle = \frac{1}{\sqrt{2}}\left(\alpha\cos\beta\,|0\rangle + \gamma \sin\beta\,|1\rangle\right).$$

For the outcome $|\Phi^-\rangle$, Bob has

$$|\tilde{b}_{\Phi^-}\rangle = \frac{1}{\sqrt{2}}\left(\gamma \sin\beta\,|0\rangle + \alpha\cos\beta\,|1\rangle\right).$$

Alice sends Bob this measurement result, and Bob applies X, giving

$$X|\tilde{b}_{\Phi^-}\rangle = \frac{1}{\sqrt{2}}\left(\alpha\cos\beta\,|0\rangle + \gamma \sin\beta\,|1\rangle\right).$$

Thus both Φ outcomes give the same corrected state,

$$|\tilde{b}_{\Phi}\rangle = \frac{1}{\sqrt{2}}\left(\alpha\cos\beta\,|0\rangle + \gamma \sin\beta\,|1\rangle\right).$$

Similarly, for the two Ψ outcomes, Alice sends Bob the result of her Bell measurement and Bob applies the corresponding Pauli correction. In both cases the corrected state is

$$|\tilde{b}_{\Psi}\rangle = \frac{1}{\sqrt{2}}\left(\alpha\sin\beta\,|0\rangle + \gamma \cos\beta\,|1\rangle\right).$$

Bob's final density matrix is obtained by summing over all four possible Bell-measurement outcomes. Since there are two Φ outcomes and two Ψ outcomes,

$$\rho_B = 2|\tilde{b}_{\Phi}\rangle\langle\tilde{b}_{\Phi}| + 2|\tilde{b}_{\Psi}\rangle\langle\tilde{b}_{\Psi}|.$$

For the two Φ outcomes,

$$2|\tilde{b}_{\Phi}\rangle\langle\tilde{b}_{\Phi}| = \begin{pmatrix} |\alpha|^2\cos^2\beta & \alpha\gamma^*\cos\beta\sin\beta \\ \alpha^*\gamma\cos\beta\sin\beta & |\gamma|^2\sin^2\beta \end{pmatrix}.$$

For the two Ψ outcomes,

$$2|\tilde{b}_\Psi\rangle\langle\tilde{b}_\Psi| = \begin{pmatrix} |\alpha|^2 \sin^2\beta & \alpha\gamma^* \sin\beta\cos\beta \\ \alpha^*\gamma \sin\beta\cos\beta & |\gamma|^2\cos^2\beta \end{pmatrix}.$$

Therefore

$$\rho_B = \begin{pmatrix} |\alpha|^2 & \alpha\gamma^* \sin 2\beta \\ \alpha^*\gamma \sin 2\beta & |\gamma|^2 \end{pmatrix}.$$

The fidelity of Bob's final state with Alice's original state is

$$F = \langle a|\rho_B|a\rangle.$$

Using

$$|a\rangle = \alpha|0\rangle + \gamma|1\rangle,$$

we obtain

$$F = |\alpha|^4 + |\gamma|^4 + 2|\alpha|^2|\gamma|^2 \sin 2\beta.$$

Now

$$|\alpha|^2 = \cos^2\left(\frac{\theta}{2}\right), \qquad |\gamma|^2 = \sin^2\left(\frac{\theta}{2}\right),$$

so

$$F = \cos^4\left(\frac{\theta}{2}\right) + \sin^4\left(\frac{\theta}{2}\right) + 2\cos^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{\theta}{2}\right)\sin 2\beta.$$

Hence

$$\boxed{F(\theta,\phi) = 1 - \frac{1}{2}\sin^2\theta\,(1-\sin 2\beta)}.$$

The fidelity is independent of ϕ. For $\beta = \pi/4$, the shared state is maximally entangled and $\sin 2\beta = 1$, so

$$F = 1.$$

1.4.3 Part C

From part (b), the fidelity for an input state specified by (θ,ϕ) is

$$F(\theta,\phi) = 1 - \frac{1}{2}\sin^2\theta\,(1-\sin 2\beta).$$

Since the unknown qubit may lie anywhere on the Bloch sphere, the average fidelity is

$$\overline{F} = \frac{1}{4\pi} \int_0^{2\pi} \int_0^{\pi} F(\theta, \phi)\, \sin\theta \, d\theta \, d\phi.$$

Substituting the expression for the fidelity,

$$\overline{F} = \frac{1}{4\pi} \int_0^{2\pi} \int_0^{\pi} \left[1 - \frac{1}{2} \sin^2\theta \, (1 - \sin 2\beta) \right] \sin\theta \, d\theta \, d\phi.$$

Since the fidelity is independent of ϕ,

$$\overline{F} = \frac{1}{2} \int_0^{\pi} \left[1 - \frac{1}{2} \sin^2\theta \, (1 - \sin 2\beta) \right] \sin\theta \, d\theta.$$

Using

$$\int_0^{\pi} \sin\theta \, d\theta = 2, \qquad \int_0^{\pi} \sin^3\theta \, d\theta = \frac{4}{3},$$

we obtain

$$\overline{F} = 1 - \frac{1}{3} \left(1 - \sin 2\beta \right).$$

Therefore,

$$\boxed{\overline{F} = \frac{2 + \sin 2\beta}{3}}.$$

For the maximally entangled state, $\beta = \pi/4$, so

$$\sin 2\beta = 1,$$

and hence

$$\overline{F} = 1,$$

as expected for perfect quantum teleportation.

1.4.4 Part D

The shared Werner state is

$$\rho_W = \frac{p}{4} I \otimes I + (1-p)|\Psi^-\rangle\langle\Psi^-|.$$

Let the unknown pure state that Alice wishes to teleport be

$$|a\rangle, \qquad \rho_a = |a\rangle\langle a|.$$

Alice performs the usual Bell-basis measurement on her two qubits, sends the two-bit measurement result to Bob, and Bob applies the corresponding Pauli correction.

Since the teleportation protocol is linear in the shared density matrix, we can consider the two parts of the Werner state separately.

For the Bell-state component

$$|\Psi^-\rangle\langle\Psi^-|,$$

the teleportation is perfect, so Bob receives

$$\rho_B^{(\Psi^-)} = |a\rangle\langle a|.$$

For the completely mixed component

$$\frac{1}{4} I \otimes I,$$

there is no correlation between Alice and Bob. Therefore Bob's qubit remains completely mixed,

$$\rho_B^{(\text{mix})} = \frac{I}{2}.$$

Hence Bob's final density matrix is

$$\rho_B = (1-p)|a\rangle\langle a| + p\frac{I}{2}.$$

Since Alice's original state is pure, the fidelity is

$$F = \langle a|\rho_B|a\rangle.$$

Therefore

$$F = (1-p)\langle a|a\rangle\langle a|a\rangle + \frac{p}{2}\langle a|I|a\rangle$$
$$= (1-p) + \frac{p}{2}.$$

Thus

$$\boxed{F = 1 - \frac{p}{2}}.$$

For $p = 0$, Alice and Bob share the perfect Bell state and

$$F = 1.$$

For $p = 1$, the shared state is completely mixed and

$$F = \frac{1}{2},$$

so Bob's state contains no information about the unknown state Alice wished to teleport.

2 Assignment 2

Question 1 Part A

Code

The following Mathematica code was used to construct the master equation in the Bloch representation, solve the resulting differential equations for an arbitrary initial state, and plot the dynamics for the initial state $|0>$.

```
ClearAll["Global`*"];


sx = {{0, 1}, {1, 0}};
sy = {{0, -I}, {I, 0}};
sz = {{1, 0}, {0, -1}};
id = IdentityMatrix[2];

comm[A_, B_] := A . B - B . A;
dissipator[a_, r_] :=
  a . r . ConjugateTranspose[a] -
```

```
   1/2 (ConjugateTranspose[a] . a . r + r . ConjugateTranspose[a] .
    a);

rho[t_] := {{1/2 + z[t]/2, x[t]/2 - I y[t]/2}, {x[t]/2 + I y[t]/2,
    1/2 - z[t]/2}};
rhs[t_] := -I /2 \[CapitalOmega] comm[sx,
     rho[t]] + \[Gamma] dissipator[sz, rho[t]];

blochEqns = {x'[t] == Simplify[Tr[sx . rhs[t]]],
   y'[t] == Simplify[Tr[sy . rhs[t]]],
   z'[t] == Simplify[Tr[sz . rhs[t]]]};
blochEqns // TableForm

initialConditions = {x[0] == x0, y[0] == y0, z[0] == z0};
rhoInitial = {{1/2 + z0/2, x0/2 - I y0/2}, {x0/2 + I y0/2, 1/2 - z0
   /2}};
rhoInitial // MatrixForm

sol = DSolve[Join[blochEqns, initialConditions], {x, y, z}, t];


xSol = x[t] /. First[sol];
ySol = y[t] /. First[sol];
zSol = z[t] /. First[sol];

assumptions = {\[CapitalOmega] > \[Gamma] > 0, t >= 0,
   Element[{x0, y0, z0}, Reals]};

yOsc = ySol /. Sqrt[\[Gamma]^2 - \[CapitalOmega]^2] -> I \[Omega];
zOsc = zSol /. Sqrt[\[Gamma]^2 - \[CapitalOmega]^2] -> I \[Omega];

yOsc = ExpToTrig[yOsc];
zOsc = ExpToTrig[zOsc];

yOsc = FullSimplify[yOsc, Assumptions -> assumptions];
zOsc = FullSimplify[zOsc, Assumptions -> assumptions];

freqRules = {Sqrt[\[CapitalOmega]^2 - \[Gamma]^2] -> \[Omega], \
(\[CapitalOmega]^2 - \[Gamma]^2)^(-1/2) -> 1/\[Omega]};
blochNice = {xSol, yOsc, zOsc} /. freqRules;
```

```
50
51 blochNice // MatrixForm
52
53 plotParameters = {\[CapitalOmega] -> 1, \[Gamma] -> 0.1, \[Omega]
       ->
54     Sqrt[1^2 - 0.1^2], x0 -> 0, y0 -> 0, z0 -> 1};
55
56 blochPlot = blochNice /. plotParameters;
57
58 Plot[Evaluate[blochPlot], {t, 0, 20},
59  PlotLegends -> {"x(t)", "y(t)", "z(t)"},
60  AxesLabel -> {"t", "Bloch component"}, PlotRange -> All,
61  GridLines -> Automatic, ImageSize -> Large]
62
63
64
65 blochEvolutionPlot =
66   Plot[Evaluate[blochPlot], {t, 0, 20},
67    PlotLegends -> {"x(t)", "y(t)", "z(t)"},
68    AxesLabel -> {"t", "Bloch component"}, PlotRange -> All,
69    GridLines -> Automatic, ImageSize -> Large];
70
71 Export["bloch_evolution.pdf", blochEvolutionPlot]
72
73
74 purityParameters = {\[CapitalOmega] -> 1, \[Gamma] -> 0.1, \[Omega]
       ->
75      Sqrt[1^2 - 0.1^2], x0 -> 0, y0 -> 0, z0 -> 1};
76 {xPlot, yPlot, zPlot} = blochNice /. purityParameters;
77 rhoPlot[t_] := {{1/2 + zPlot/2,
78     xPlot/2 - I yPlot/2}, {xPlot/2 + I yPlot/2, 1/2 - zPlot/2}};
79 purity[t_] := FullSimplify[Tr[rhoPlot[t] . rhoPlot[t]]];
80 purity[t]
81
82 purityPlot =
83   Plot[Evaluate[purity[t]], {t, 0, 20},
84    AxesLabel -> {Style["t", 14], Style["Tr[\[Rho](t)^2]", 14]},
85    PlotRange -> {0, 1}, GridLines -> Automatic, ImageSize -> Large,
86    BaseStyle -> {FontSize -> 12}];
87
```

```
purityPlot
Export["purity_evolution.pdf", purityPlot]
```

1. Dynamics of the qubit

The state of the qubit can be written in terms of its Bloch vector as

$$\rho(t) = \frac{1}{2}\left(I + x(t)\sigma_x + y(t)\sigma_y + z(t)\sigma_z\right),$$

or equivalently,

$$\rho(t) = \begin{pmatrix} \dfrac{1+z(t)}{2} & \dfrac{x(t)-iy(t)}{2} \\ \dfrac{x(t)+iy(t)}{2} & \dfrac{1-z(t)}{2} \end{pmatrix}.$$

Substituting this into the master equation gives the coupled differential equations

$$\boxed{\begin{aligned} \dot{x}(t) &= -2\gamma x(t), \\ \dot{y}(t) &= -2\gamma y(t) - \Omega z(t), \\ \dot{z}(t) &= \Omega y(t). \end{aligned}}$$

For an arbitrary initial Bloch vector

$$\mathbf{r}(0) = \begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix},$$

the initial density matrix is therefore

$$\rho(0) = \rho_0 = \frac{1}{2}\begin{pmatrix} 1+z_0 & x_0 - iy_0 \\ x_0 + iy_0 & 1-z_0 \end{pmatrix}.$$

Defining

$$\omega = \sqrt{\Omega^2 - \gamma^2},$$

the solution of the Bloch equations is

$$\boxed{x(t) = x_0 e^{-2\gamma t}}$$

and

$$\boxed{y(t) = e^{-\gamma t}\left[y_0\cos(\omega t) - \frac{\gamma y_0 + \Omega z_0}{\omega}\sin(\omega t)\right].}$$

Similarly,

$$\boxed{z(t) = e^{-\gamma t}\left[z_0\cos(\omega t) + \frac{\gamma z_0 + \Omega y_0}{\omega}\sin(\omega t)\right].}$$

Hence, for an arbitrary initial state, the Bloch vector evolves as

$$\boxed{\mathbf{r}(t) = \begin{pmatrix} x_0 e^{-2\gamma t} \\ e^{-\gamma t}\left[y_0\cos(\omega t) - \dfrac{\gamma y_0 + \Omega z_0}{\omega}\sin(\omega t)\right] \\ e^{-\gamma t}\left[z_0\cos(\omega t) + \dfrac{\gamma z_0 + \Omega y_0}{\omega}\sin(\omega t)\right] \end{pmatrix}.}$$

The density matrix at any later time is then obtained from

$$\rho(t) = \frac{1}{2}\begin{pmatrix} 1 + z(t) & x(t) - iy(t) \\ x(t) + iy(t) & 1 - z(t) \end{pmatrix}.$$

2. Steady state

At steady state, the components of the Bloch vector no longer change with time, so

$$\dot{x} = \dot{y} = \dot{z} = 0.$$

Using the Bloch equations found above,

$$0 = -2\gamma x,$$

$$0 = -2\gamma y - \Omega z,$$

$$0 = \Omega y.$$

For $\gamma > 0$ and $\Omega \neq 0$, these equations give

$$x_{\text{ss}} = 0, \qquad y_{\text{ss}} = 0, \qquad z_{\text{ss}} = 0.$$

Therefore, the steady-state Bloch vector is

$$\mathbf{r}_{\text{ss}} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Substituting this into the density matrix gives

$$\rho_{\text{ss}} = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \boxed{\frac{1}{2}I}.$$

Thus, the steady state is the maximally mixed state.

3. Dependence on the initial condition

The steady state found above is

$$\rho_{\text{ss}} = \frac{1}{2} I,$$

and therefore does not depend on the initial state of the qubit.

This can also be seen directly from the solutions for the Bloch-vector components. Each component contains an exponentially decaying factor,

$$x(t) \propto e^{-2\gamma t}, \qquad y(t) \propto e^{-\gamma t}, \qquad z(t) \propto e^{-\gamma t}.$$

For $\gamma > 0$,

$$e^{-\gamma t} \to 0 \qquad \text{as} \qquad t \to \infty,$$

so irrespective of the initial values x_0, y_0, and z_0,

$$x(t) \to 0, \qquad y(t) \to 0, \qquad z(t) \to 0.$$

Hence,

$$\mathbf{r}(t) \to \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix},$$

and consequently

$$\rho(t) \to \frac{1}{2} I.$$

Thus, the steady state is independent of the initial condition.

4. Plot of the system as a function of time

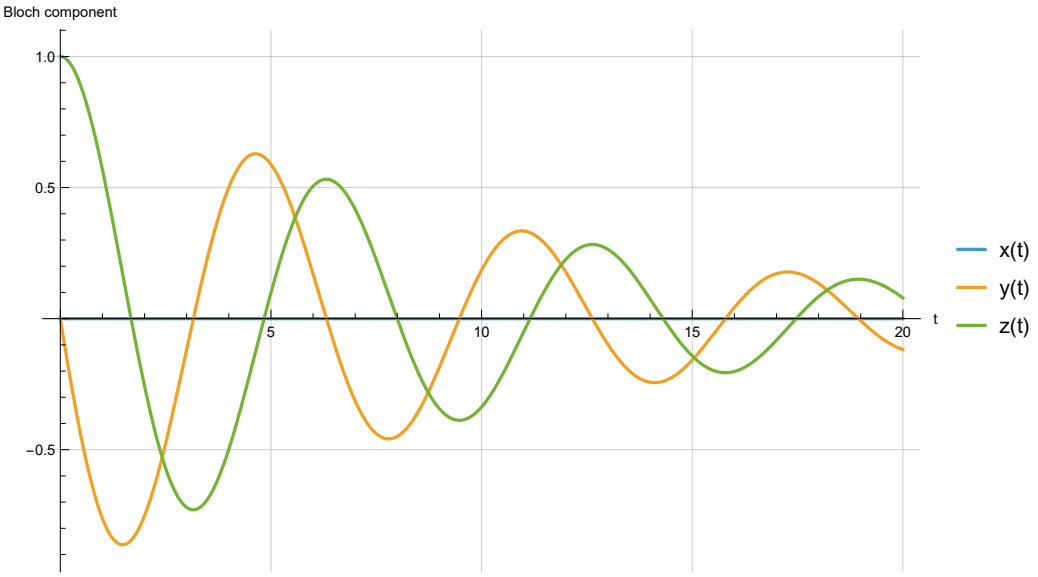


Figure 1: Evolution of the Bloch-vector components for the initial state $|0\rangle$, with $\Omega = 1$ and $\gamma = 0.1$.

5. Purity plot

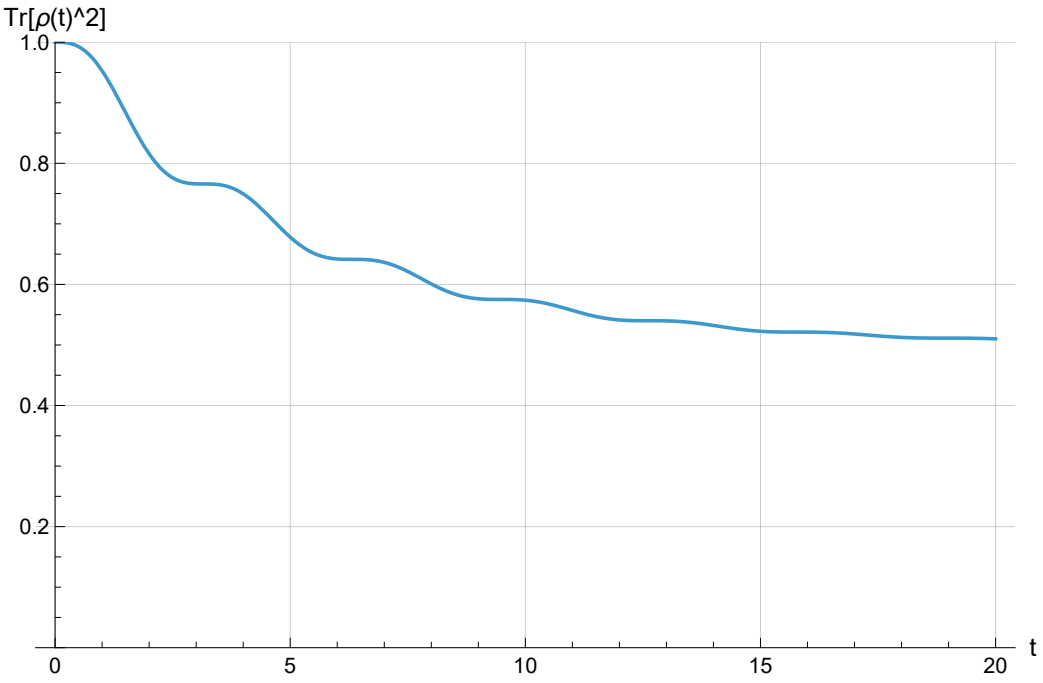


Figure 2: Evolution of the purity for the initial state $|0\rangle$, with $\Omega = 1$ and $\gamma = 0.1$.

Question 1 Part B

Code

The following Mathematica code was used to express the master equation in terms of the Bloch-vector components and solve the resulting differential equations for an arbitrary initial state. The basis convention used is $|g\rangle, |e\rangle$, where

$$|g\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |e\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

```
ClearAll["Global`*"];

sx = {{0, 1}, {1, 0}};
sy = {{0, -I}, {I, 0}};
sz = {{1, 0}, {0, -1}};
id = IdentityMatrix[2];

(* Lowering and raising operators *)
sm = {{0, 1}, {0, 0}};   (* sigma_- = |g><e| *)
sp = {{0, 0}, {1, 0}};   (* sigma_+ = |e><g| *)

comm[A_, B_] := A . B - B . A;

dissipator[a_, r_] :=
  a . r . ConjugateTranspose[a] -
   1/2 (
     ConjugateTranspose[a] . a . r +
     r . ConjugateTranspose[a] . a
     );

(* Density matrix in Bloch-vector form *)
rho[t_] := {
   {1/2 + z[t]/2, x[t]/2 - I y[t]/2},
   {x[t]/2 + I y[t]/2, 1/2 - z[t]/2}
   };

(* Master equation *)
rhs[t_] :=
  \[Gamma] (n + 1)/2 dissipator[sm, rho[t]] +
   \[Gamma] n/2 dissipator[sp, rho[t]];
```

```
(* Bloch equations *)
blochEqns = {
   x'[t] == Simplify[Tr[sx . rhs[t]]],
   y'[t] == Simplify[Tr[sy . rhs[t]]],
   z'[t] == Simplify[Tr[sz . rhs[t]]]
   };

blochEqns // TableForm

(* Arbitrary initial state *)
initialConditions = {
   x[0] == x0,
   y[0] == y0,
   z[0] == z0
   };

rhoInitial = {
   {1/2 + z0/2, x0/2 - I y0/2},
   {x0/2 + I y0/2, 1/2 - z0/2}
   };

rhoInitial // MatrixForm

(* Solve the Bloch equations *)
sol = DSolve[
   Join[blochEqns, initialConditions],
   {x, y, z},
   t
   ];

xSol = x[t] /. First[sol];
ySol = y[t] /. First[sol];
zSol = z[t] /. First[sol];

assumptions = {
   n >= 0,
   \[Gamma] > 0,
   t >= 0,
   Element[{x0, y0, z0}, Reals]
```

```
   };

xSol = FullSimplify[xSol, Assumptions -> assumptions];
ySol = FullSimplify[ySol, Assumptions -> assumptions];
zSol = FullSimplify[zSol, Assumptions -> assumptions];

blochNice = {xSol, ySol, zSol};

blochNice // MatrixForm

(* Example: zero-temperature decay from |e> *)
plotParameters = {
   n -> 0,
   \[Gamma] -> 0.1,
   x0 -> 0,
   y0 -> 0,
   z0 -> -1
   };

blochPlot = blochNice /. plotParameters;

Plot[
 Evaluate[blochPlot],
 {t, 0, 20},
 PlotLegends -> {"x(t)", "y(t)", "z(t)"},
 AxesLabel -> {"t", "Bloch component"},
 PlotRange -> All,
 GridLines -> Automatic,
 ImageSize -> Large
 ]
```

a) General Solution

For an arbitrary initial Bloch vector

$$\mathbf{r}(0) = \begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix},$$

the solution is

$$\boxed{x(t) = x_0 \exp\left[-\frac{\gamma(1+2n)t}{4}\right]}$$

and

$$\boxed{y(t) = y_0 \exp\left[-\frac{\gamma(1+2n)t}{4}\right].}$$

For the z-component,

$$\boxed{z(t) = \frac{1 + \exp\left[-\frac{\gamma(1+2n)t}{2}\right][-1 + (1+2n)z_0]}{1+2n}.}$$

Equivalently, the Bloch vector may be written as

$$\boxed{\mathbf{r}(t) = \begin{pmatrix} x_0 e^{-\gamma(1+2n)t/4} \\ y_0 e^{-\gamma(1+2n)t/4} \\ \dfrac{1 + e^{-\gamma(1+2n)t/2}[-1 + (1+2n)z_0]}{1+2n} \end{pmatrix}.}$$

b) Steady state

The solution for the z-component can be written in the form

$$z(t) = \frac{1}{1+2n} + \left(z_0 - \frac{1}{1+2n} \right) e^{-\gamma(1+2n)t/2}.$$

Similarly,

$$x(t) = x_0 e^{-\gamma(1+2n)t/4}, \qquad y(t) = y_0 e^{-\gamma(1+2n)t/4}.$$

Since $\gamma > 0$ and $n \geq 0$,

$$e^{-\gamma(1+2n)t/4} \to 0, \qquad e^{-\gamma(1+2n)t/2} \to 0$$

as $t \to \infty$. Therefore,

$$x_{\text{ss}} = 0, \qquad y_{\text{ss}} = 0, \qquad z_{\text{ss}} = \frac{1}{1+2n}.$$

Hence the steady-state Bloch vector is

$$\mathbf{r}_{\text{ss}} = \begin{pmatrix} 0 \\ 0 \\ \dfrac{1}{1+2n} \end{pmatrix}.$$

Using

$$\rho = \frac{1}{2} \begin{pmatrix} 1+z & x-iy \\ x+iy & 1-z \end{pmatrix},$$

the steady-state density matrix is

$$\rho_{\text{ss}} = \frac{1}{2} \begin{pmatrix} 1 + \dfrac{1}{1+2n} & 0 \\ 0 & 1 - \dfrac{1}{1+2n} \end{pmatrix}.$$

This simplifies to

$$\boxed{\rho_{\text{ss}} = \begin{pmatrix} \dfrac{n+1}{2n+1} & 0 \\ 0 & \dfrac{n}{2n+1} \end{pmatrix}.}$$

Since the steady state depends only on n and not on the initial Bloch-vector components x_0, y_0, or z_0, it is independent of the initial state.

c) Kraus operator-sum representation

The evolution generated by this master equation corresponds to a generalised amplitude-damping channel. Let

$$\Delta t = t_f - t_i$$

be the total evolution time, and define

$$\eta = \exp\left[-\frac{\gamma(2n+1)\Delta t}{2}\right].$$

From the solution obtained in part (a), the off-diagonal elements of the density matrix decay as

$$\rho_{ge}(t_f) = \sqrt{\eta}\,\rho_{ge}(t_i),$$

while the populations relax towards the steady-state populations

$$p_g = \frac{n+1}{2n+1}, \qquad p_e = \frac{n}{2n+1}.$$

Defining

$$p = \frac{n+1}{2n+1}, \qquad 1-p = \frac{n}{2n+1},$$

and

$$\lambda = 1 - \eta = 1 - \exp\left[-\frac{\gamma(2n+1)\Delta t}{2}\right],$$

the total evolution can be written in Kraus operator-sum form as

$$\boxed{\rho(t_f) = \sum_{k=0}^{3} K_k\,\rho(t_i)\,K_k^{\dagger}.}$$

Using the basis

$$|g\rangle, |e\rangle$$

a set of Kraus operators is

$$\boxed{K_0 = \sqrt{p}\begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\lambda} \end{pmatrix},}$$

$$\boxed{K_1 = \sqrt{p}\begin{pmatrix} 0 & \sqrt{\lambda} \\ 0 & 0 \end{pmatrix},}$$

$$\boxed{K_2 = \sqrt{1-p}\begin{pmatrix} \sqrt{1-\lambda} & 0 \\ 0 & 1 \end{pmatrix},}$$

and

$$\boxed{K_3 = \sqrt{1-p}\begin{pmatrix} 0 & 0 \\ \sqrt{\lambda} & 0 \end{pmatrix}.}$$

These operators satisfy the completeness relation

$$\sum_{k=0}^{3} K_k^\dagger K_k = I,$$

and therefore define a trace-preserving quantum channel.

Substituting the values of p and λ gives

$$p = \frac{n+1}{2n+1}, \qquad \lambda = 1 - e^{-\gamma(2n+1)\Delta t/2}.$$

In the long-time limit,

$$\Delta t \to \infty \quad \Longrightarrow \quad \lambda \to 1,$$

and the channel maps every initial state to

$$\rho_{\rm ss} = \begin{pmatrix} \dfrac{n+1}{2n+1} & 0 \\ 0 & \dfrac{n}{2n+1} \end{pmatrix},$$

in agreement with the steady state obtained in part (b).

For $n = 0$, $p = 1$, so K_2 and K_3 vanish and the channel reduces to the ordinary amplitude-damping channel describing decay from $|e\rangle$ to $|g\rangle$.

Equivalently, the Kraus operators may be written directly in terms of n, γ, and the total evolution time

$$\Delta t = t_f - t_i.$$

Since

$$p = \frac{n+1}{2n+1}, \qquad 1 - p = \frac{n}{2n+1},$$

and

$$1-\lambda = \exp\left[-\frac{\gamma(2n+1)\Delta t}{2}\right],$$

we have

$$\sqrt{1-\lambda} = \exp\left[-\frac{\gamma(2n+1)\Delta t}{4}\right],$$

and

$$\sqrt{\lambda} = \sqrt{1-\exp\left[-\frac{\gamma(2n+1)\Delta t}{2}\right]}.$$

Therefore,

$$\boxed{K_0 = \sqrt{\frac{n+1}{2n+1}}\begin{pmatrix}1 & 0\\ 0 & e^{-\gamma(2n+1)\Delta t/4}\end{pmatrix}}$$

$$\boxed{K_1 = \sqrt{\frac{n+1}{2n+1}}\begin{pmatrix}0 & \sqrt{1-e^{-\gamma(2n+1)\Delta t/2}}\\ 0 & 0\end{pmatrix}}$$

$$\boxed{K_2 = \sqrt{\frac{n}{2n+1}}\begin{pmatrix}e^{-\gamma(2n+1)\Delta t/4} & 0\\ 0 & 1\end{pmatrix}}$$

and

$$\boxed{K_3 = \sqrt{\frac{n}{2n+1}}\begin{pmatrix}0 & 0\\ \sqrt{1-e^{-\gamma(2n+1)\Delta t/2}} & 0\end{pmatrix}}.$$

Hence the total evolution from t_i to t_f is

$$\boxed{\rho(t_f) = \sum_{k=0}^{3} K_k\,\rho(t_i)\,K_k^\dagger,}$$

with the four Kraus operators given above.

Question 2: The Steane Code

1. Number of logical qubits

For a stabiliser code containing n physical qubits and r independent stabiliser generators, the number of encoded logical qubits is

$$k = n - r.$$

There are seven physical qubits and six independent stabiliser generators, so

$$k = 7 - 6 = 1.$$

Therefore, the Steane code encodes

$$\boxed{k = 1}$$

logical qubit, and has the form

$$[[7, 1, d]].$$

2. Tanner graph and commutation of the stabilisers

The Tanner graph is shown below. Since the Z-type stabilisers have exactly the same support, their Tanner graph is identical.

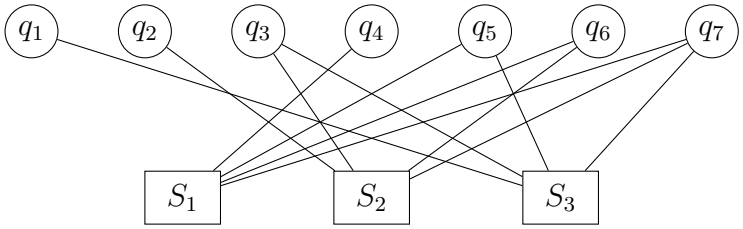


Figure 3: Tanner graph for the three stabiliser supports of the Steane code. The X-type and Z-type stabilisers have the same graph.

All X-type stabilisers commute with each other, since they contain only I and X. Similarly, all Z-type stabilisers commute with each other.

It remains to check the commutation between the X-type and Z-type stabilisers. Since

$$XZ = -ZX,$$

two Pauli strings commute if the number of positions on which one contains X and the other contains Z is even.

Each X-type stabiliser overlaps with its corresponding Z-type stabiliser on four qubits. For example,

$$S_1^X = IIIXXXX, \qquad S_1^Z = IIIZZZZ,$$

so there are four XZ overlaps and therefore

$$(-1)^4 = +1.$$

Different stabiliser generators overlap on two qubits. For example,

$$S_1^X = IIIXXXX, \qquad S_2^Z = IZZIIZZ,$$

overlap on qubits 6 and 7, giving

$$(-1)^2 = +1.$$

3. Logical operators and code distance

A convenient choice of logical Pauli operators is initially

$$X_L = XXXXXXX, \qquad Z_L = ZZZZZZZ.$$

Both operators commute with all stabilisers because they overlap each opposite-type stabiliser on four qubits.

However, the logical operators mutually anticommute. Since X_L and Z_L overlap on all seven qubits,

$$X_L Z_L = (-1)^7 Z_L X_L = -Z_L X_L.$$

Logical operators may be multiplied by stabilisers without changing their logical action. For example,

$$X_L S_1^X = (XXXXXXX)(IIIXXXX) = XXXIIII.$$

Therefore, an equivalent logical X operator is

$$\boxed{X_L = XXXIIII.}$$

Similarly,

$$Z_L S_1^Z = (ZZZZZZZ)(IIIZZZZ) = ZZZIIII,$$

so an equivalent logical Z operator is

$$\boxed{Z_L = ZZZIIII.}$$

These operators still anticommute, since they overlap on three qubits:

$$(-1)^3 = -1.$$

To see that no lower-weight logical operator exists, consider the columns of the parity-check matrix:

Qubit	Syndrome
q_1	001
q_2	010
q_3	011
q_4	100
q_5	101
q_6	110
q_7	111

.

Every column is non-zero, so every single-qubit error produces a non-zero syndrome. Furthermore, all columns are distinct, so all single-qubit errors can be distinguished.

Thus there is no non-trivial logical operator of weight one or two. Since a weight-three logical operator exists, the minimum logical weight is

$$\boxed{d = 3}.$$

Hence the Steane code is the

$$\boxed{[[7, 1, 3]]}$$

quantum error-correcting code.

4. Lookup-table decoder

Assume an independent X–Z error model, in which each physical qubit independently experiences an X error component with probability p_X and a Z error component with probability p_Z.

Since the Steane code is a CSS code, the X and Z error components may be decoded separately.

An X_i error anticommutes with the Z-type stabilisers that have support on qubit i. Measurement of the three Z-type stabilisers therefore produces a three-bit syndrome identifying a single X error:

Syndrome	Correction
000	I
001	X_1
010	X_2
011	X_3
100	X_4
101	X_5
110	X_6
111	X_7

.

Similarly, a Z_i error is identified by measuring the three X-type stabilisers. Because the Steane code is self-dual, the same lookup table can be used:

Syndrome	Correction
000	I
001	Z_1
010	Z_2
011	Z_3
100	Z_4
101	Z_5
110	Z_6
111	Z_7

.

The decoder is guaranteed to succeed whenever there are at most one X-error component and at most one Z-error component.

For the X sector, the probability of zero or one X errors is

$$P_{\le 1}^{(X)} = (1-p_X)^7 + 7p_X(1-p_X)^6.$$

Hence the probability of two or more X errors is

$$P_{\geq 2}^{(X)} = 1 - (1 - p_X)^7 - 7p_X(1 - p_X)^6.$$

Similarly,

$$P_{\geq 2}^{(Z)} = 1 - (1 - p_Z)^7 - 7p_Z(1 - p_Z)^6.$$

A simple union bound on the decoder failure probability is therefore

$$\boxed{P_{\text{fail}} \leq P_{\geq 2}^{(X)} + P_{\geq 2}^{(Z)}.}$$

That is,

$$\boxed{\begin{aligned} P_{\text{fail}} \leq\ & 1 - (1 - p_X)^7 - 7p_X(1 - p_X)^6 \\ & + 1 - (1 - p_Z)^7 - 7p_Z(1 - p_Z)^6. \end{aligned}}$$

Since the X and Z error processes are assumed to be independent, a tighter bound is obtained from the probability that both sectors contain at most one error:

$$\boxed{P_{\text{fail}} \leq 1 - \left[(1 - p_X)^7 + 7p_X(1 - p_X)^6\right]\left[(1 - p_Z)^7 + 7p_Z(1 - p_Z)^6\right].}$$

5. Correction of a single-qubit Y error

A Pauli Y error can be written as

$$Y = iXZ.$$

Therefore, a Y_i error on physical qubit i contains both an X_i component and a Z_i component.

The X_i component is detected by measurement of the Z-type stabilisers, while the Z_i component is detected independently by measurement of the X-type stabilisers.

For example, if a Y error occurs on qubit i, both syndrome measurements identify the same physical qubit. The decoder therefore identifies

$$X_i \qquad \text{and} \qquad Z_i$$

and applies the corresponding corrections.

Since

$$Z_i X_i = -iY_i,$$

the resulting operator differs from Y_i only by a physically irrelevant global phase. Thus the single-qubit Y error is corrected.

This result can also be checked using the quantum error-correction criteria. A set of errors E_a is correctable if

$$\boxed{\langle i_L | E_a^\dagger E_b | j_L \rangle = C_{ab}\delta_{ij},}$$

for all logical basis states $|i_L\rangle$ and $|j_L\rangle$.

For the Steane code, consider the set of all single-qubit Pauli errors,

$$\mathcal{E} = \{I, X_i, Y_i, Z_i \,:\, i = 1, \ldots, 7\}.$$

For any two errors $E_a, E_b \in \mathcal{E}$, the operator

$$E_a^\dagger E_b$$

has weight at most two. Since the Steane code has distance

$$d = 3,$$

no Pauli operator of weight one or two can act as a non-trivial logical operator.

Therefore, for two different logical basis states,

$$\langle 0_L|E_a^\dagger E_b|1_L\rangle = 0.$$

This means that the error and subsequent syndrome measurement do not mix the logical basis states $|0_L\rangle$ and $|1_L\rangle$.

The second requirement is that

$$\langle 0_L|E_a^\dagger E_b|0_L\rangle = \langle 1_L|E_a^\dagger E_b|1_L\rangle.$$

This is important because it means that the error syndrome contains no information about whether the encoded state was $|0_L\rangle$ or $|1_L\rangle$. Therefore, measuring the error syndrome does not reveal the state of the logical qubit.

For example, if the encoded state is an arbitrary superposition

$$|\psi_L\rangle = \alpha|0_L\rangle + \beta|1_L\rangle,$$

the syndrome measurement can determine which physical error occurred without determining the values of α and β, or distinguishing between the logical components $|0_L\rangle$ and $|1_L\rangle$. Hence the logical superposition is not destroyed by the error-correction process.

For the particular case of a single Y_i error,

$$Y_i^\dagger Y_i = I,$$

so

$$\langle 0_L|Y_i^\dagger Y_i|0_L\rangle = \langle 1_L|Y_i^\dagger Y_i|1_L\rangle = 1,$$

while

$$\langle 0_L|Y_i^\dagger Y_i|1_L\rangle = \langle 0_L|1_L\rangle = 0.$$

Thus a single-qubit Y error satisfies the quantum error-correction criteria and is correctable by the Steane code.

Practically, the decoder uses

$$\boxed{Y_i = iX_iZ_i}$$

and treats the Y_i error as simultaneous X_i and Z_i error components. The Z-type stabilisers identify the X_i component, while the X-type stabilisers identify the Z_i component.

Part B

a) Failure rate for bit-flip errors

Assume that each of the seven physical qubits independently experiences a bit-flip error with probability p, with no phase-flip errors.

The lookup-table decoder succeeds whenever the actual error differs from the correction chosen by the decoder only by an X-type stabiliser. Thus a successfully corrected error can be written as

$$E = CS,$$

where

$$C \in \{I, X_1, \ldots, X_7\}$$

is the correction chosen from the lookup table, and

$$S \in \mathcal{S}_X$$

is an element of the X-type stabiliser group.

The X-type stabiliser group contains eight elements:

$$\mathcal{S}_X = \{I,\ \text{seven weight-4 stabilisers}\}.$$

The successfully corrected error patterns can therefore be counted as follows.

For $C = I$ and $S = I$, there is one successful weight-zero pattern.

For $C = X_i$ and $S = I$, there are seven successful weight-one patterns.

Each non-trivial X-type stabiliser has weight four, so there are 7 successful weight-four patterns.

Multiplying a weight-four stabiliser by a single-qubit correction X_i gives two possibilities. If X_i acts on one of the four qubits already contained in the stabiliser, one X cancels and the resulting operator has weight three. Hence there are

$$7 \times 4 = 28$$

successful weight-three patterns.

If instead X_i acts on one of the three qubits not contained in the stabiliser, an additional X is introduced and the resulting operator has weight five. Hence there are

$$7 \times 3 = 21$$

successful weight-five patterns.

Therefore,

w	0	1	2	3	4	5	6	7
N_{success}	1	7	0	28	7	21	0	0

.

The total number of possible error patterns of weight w is

$$\binom{7}{w}.$$

Hence,

$$N_{\text{fail}}(w) = \binom{7}{w} - N_{\text{success}}(w),$$

giving

w	0	1	2	3	4	5	6	7
$\binom{7}{w}$	1	7	21	35	35	21	7	1
N_{fail}	0	0	21	7	28	0	7	1

.

An error pattern of weight w occurs with probability

$$p^w(1-p)^{7-w}.$$

Therefore, the exact failure probability of the lookup-table decoder is

$$\boxed{\begin{aligned} P_{\text{fail}}(p) = {} & 21p^2(1-p)^5 + 7p^3(1-p)^4 \\ & + 28p^4(1-p)^3 + 7p^6(1-p) + p^7. \end{aligned}}$$

2.0.1 b) Transversal CNOT

Consider two Steane-code blocks, with the first logical qubit acting as the control and the second as the target. Apply seven physical CNOT gates between corresponding physical qubits:

$$U = \prod_{i=1}^{7} \mathrm{CNOT}_{C_i \to T_i} .$$

Thus,

$$C_1 \to T_1, \quad C_2 \to T_2, \quad \ldots, \quad C_7 \to T_7.$$

Under a physical CNOT, the Pauli operators transform in the Heisenberg picture as

$$X_C \to X_C X_T, \qquad Z_C \to Z_C,$$
$$X_T \to X_T, \qquad Z_T \to Z_C Z_T.$$

Since the Steane code has identical X- and Z-type stabiliser patterns in each block, the stabilisers transform as

$$S_i^{X,C} \to S_i^{X,C} S_i^{X,T}, \qquad S_i^{X,T} \to S_i^{X,T},$$

and

$$S_i^{Z,C} \to S_i^{Z,C}, \qquad S_i^{Z,T} \to S_i^{Z,C} S_i^{Z,T}.$$

Each transformed operator is therefore still a stabiliser or a product of stabilisers, so the operation preserves the encoded code space.

Now consider the logical Pauli operators. Choosing

$$X_L = XXXXXXX, \qquad Z_L = ZZZZZZZ,$$

the transversal CNOT gives

$$X_L^{(C)} \to X_L^{(C)} X_L^{(T)},$$
$$Z_L^{(C)} \to Z_L^{(C)},$$
$$X_L^{(T)} \to X_L^{(T)},$$

and

$$Z_L^{(T)} \to Z_L^{(C)} Z_L^{(T)}.$$

These are exactly the transformation rules for a CNOT gate acting on two logical qubits. Therefore,

$$\boxed{\mathrm{CNOT}_L = \prod_{i=1}^{7} \mathrm{CNOT}_{C_i \to T_i}\,.}$$

Hence the logical CNOT is transversal and can be implemented by seven independent physical CNOT gates between corresponding qubits in the two Steane-code blocks.

Assignment 3

Question 1

2.1 Single-qubit rotation about the x-axis

The unitary operator describing a rotation through an angle θ about the x-axis is

$$U_x(\theta) = e^{-i\theta\sigma_x/2}.$$

Using the identity

$$e^{-i\theta\sigma_x/2} = \cos\left(\frac{\theta}{2}\right) I - i \sin\left(\frac{\theta}{2}\right) \sigma_x,$$

and substituting

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

we obtain

$$U_x(\theta) = \cos\left(\frac{\theta}{2}\right) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - i \sin\left(\frac{\theta}{2}\right) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Therefore,

$$\boxed{U_x(\theta) = \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) & -i\sin\left(\frac{\theta}{2}\right) \\ -i\sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{pmatrix}}$$

Hence the full action on the two qubit state is:

$$\boxed{U_{\mathrm{CU}_x}(\theta) = \begin{pmatrix} \cos\theta/2 & -i\sin\theta/2 & 0 & 0 \\ -i\sin\theta/2 & \cos\theta/2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}}$$

2.2 Two Qubit Gate Implementation

The gate can be implemented using two adjacent trapped atoms, exploiting the Rydberg–Rydberg blockade interaction. A resonant laser pulse is applied selectively to the target atom, coupling its ground state $|g\rangle_t$ and Rydberg state $|r\rangle_t$ with Rabi frequency Ω.

When the control atom is in its ground state $|g\rangle_c$, there is no significant Rydberg–Rydberg interaction, and the target undergoes ordinary Rabi oscillations. By selecting the pulse duration τ, a rotation of

$$\boxed{\theta = \Omega\tau}$$

can be achieved.

However, when the control atom is in its Rydberg state $|r\rangle_c$, the Rydberg–Rydberg interaction shifts the energy of the doubly excited state $|rr\rangle$ by Δ_{rr}. This detunes the target transition from the applied laser frequency, suppressing Rabi oscillations through the Rydberg blockade mechanism.

Therefore, the target undergoes the desired $U_x(\theta)$ rotation when the control atom is in $|g\rangle_c$, but remains approximately unchanged when the control atom is in $|r\rangle_c$, implementing the controlled rotation gate.

2.3 Pulse response

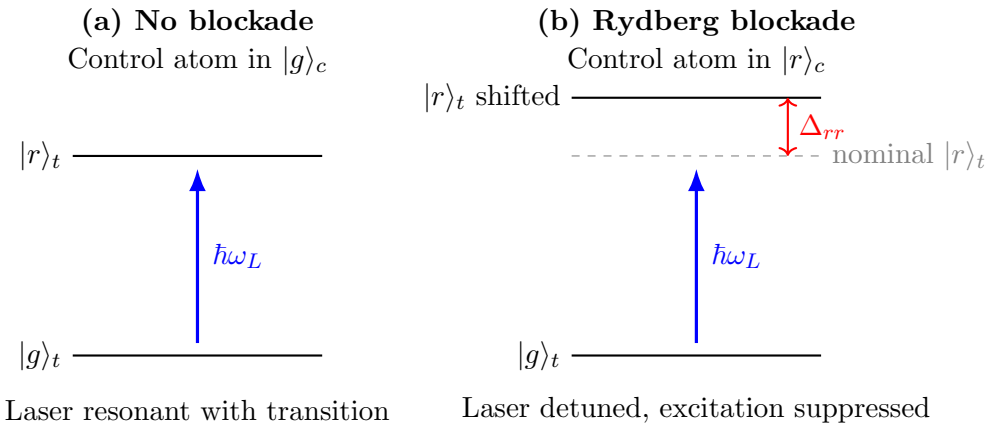


Figure 4: Energy-level picture of the controlled rotation using the Rydberg blockade mechanism. In the absence of blockade, the applied pulse is resonant with the target transition. When the control atom is already in the Rydberg state, the target Rydberg level is shifted by the interaction energy Δ_{rr}, so the same pulse is off-resonant and the rotation is suppressed.

2.4 Rydberg Excitation Shift

For the gate to operate with high fidelity, the Rydberg–Rydberg interaction shift Δ_{rr} must be sufficiently large compared with the Rabi frequency Ω, such that

$$\boxed{\frac{|\Delta_{rr}|}{\hbar} \gg \Omega}$$

where Δ_{rr} is the energy shift of the doubly excited state $|rr\rangle$.

When the control atom is in its Rydberg state $|r\rangle_c$, this interaction shifts the target transition away from resonance with the applied laser. The corresponding detuning is $\delta = \Delta_{rr}/\hbar$.

2.5 Scalability

A significant scalability problem arises because the Rydberg–Rydberg blockade interaction is not confined to the intended control and target atoms. In a

larger array, other nearby atoms occupying Rydberg states may also interact with the target, causing unwanted energy shifts and suppressing transitions that should otherwise occur.

Since the interaction strength decreases with atomic separation, increasing the spacing between atoms can reduce unwanted interactions, but also weakens the blockade required for the intended gate. This creates a trade-off between gate fidelity and unwanted interactions in larger systems.

Furthermore, because the computational basis includes a Rydberg state, qubits are susceptible to spontaneous decay and decoherence, which becomes increasingly problematic as the system scales.

2.6 CNOT Gate

To construct a CNOT gate, we require the controlled rotation to implement a Pauli X operation on the target qubit, while ensuring that no relative phase is introduced between the control states.

We first choose the rotation angle $\theta = 3\pi$, giving

$$U_x(3\pi) = \begin{pmatrix} \cos(3\pi/2) & -i\sin(3\pi/2) \\ -i\sin(3\pi/2) & \cos(3\pi/2) \end{pmatrix} = iX.$$

The corresponding controlled rotation is therefore

$$U_{\mathrm{CU}_x}(3\pi) = \begin{pmatrix} 0 & i & 0 & 0 \\ i & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

The factor of i applies only when the control qubit is in $|0\rangle_c$, introducing an unwanted relative phase.

To correct this, we first apply a z-rotation of $\pi/2$ to the control qubit:

$$R_z\left(\frac{\pi}{2}\right) \otimes I = \begin{pmatrix} e^{-i\pi/4} & 0 & 0 & 0 \\ 0 & e^{-i\pi/4} & 0 & 0 \\ 0 & 0 & e^{i\pi/4} & 0 \\ 0 & 0 & 0 & e^{i\pi/4} \end{pmatrix}.$$

The combined operation is

$$
\begin{aligned}
U_{\text{total}} &= U_{\text{CU}_x}(3\pi)\left(R_z\left(\frac{\pi}{2}\right)\otimes I\right) \\
&= \begin{pmatrix} 0 & ie^{-i\pi/4} & 0 & 0 \\ ie^{-i\pi/4} & 0 & 0 & 0 \\ 0 & 0 & e^{i\pi/4} & 0 \\ 0 & 0 & 0 & e^{i\pi/4} \end{pmatrix} \\
&= e^{i\pi/4}\begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.
\end{aligned}
$$

Since $e^{i\pi/4}$ is a global phase, it has no observable effect. The resulting gate is therefore a zero-controlled CNOT, which flips the target when the control is in $|0\rangle_c$.

Finally, the required pulse duration follows from $\theta = \Omega\tau$:

$$\boxed{\tau = \frac{3\pi}{\Omega}.}$$

A conventional CNOT (controlled on $|1\rangle_c$) can be obtained by applying an X gate to the control qubit before and after this operation.

2.7 Pulse Time

From Question 1.6, the required rotation angle is $\theta = 3\pi$. Using $\theta = \Omega\tau$, and $\Omega = 2\pi(2\times10^3)\,\text{rad}\,\text{s}^{-1}$,

$$\tau = \frac{3\pi}{\Omega} = \frac{3\pi}{2\pi(2000)} = 7.5\times10^{-4}\,\text{s}.$$

Therefore, the required pulse duration is

$$\boxed{\tau = 0.75\,\text{ms}.}$$

Question 2.1

The first pulse is a red sideband π-pulse applied to ion A. This couples the electronic and motional states

$$|e,0\rangle_A \longleftrightarrow |g,1\rangle_A ,$$

where the second quantum number represents the number of phonons in the collective motional mode.

A rotation about the x-axis is given by

$$U_x(\theta) = e^{-i\theta\sigma_x/2} = \begin{pmatrix} \cos(\theta/2) & -i\sin(\theta/2) \\ -i\sin(\theta/2) & \cos(\theta/2) \end{pmatrix}.$$

For a π-pulse, $\theta = \pi$, giving

$$U_x(\pi) = \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix} = -i\sigma_x.$$

Therefore, within the two-state subspace coupled by the red sideband, the pulse exchanges the states with a phase factor of $-i$:

$$\begin{aligned} |e,0\rangle_A &\longrightarrow -i\,|g,1\rangle_A , \\ |g,1\rangle_A &\longrightarrow -i\,|e,0\rangle_A . \end{aligned}$$

However, $|g,0\rangle_A$ is unaffected because the red sideband transition requires the removal of a phonon when exciting the ion, and no phonons are initially present. Therefore:

$$\boxed{\begin{aligned} |gg,0\rangle &\longrightarrow |gg,0\rangle , \\ |ge,0\rangle &\longrightarrow |ge,0\rangle , \\ |eg,0\rangle &\longrightarrow -i\,|gg,1\rangle , \\ |ee,0\rangle &\longrightarrow -i\,|ge,1\rangle . \end{aligned}}$$

Question 2.2

The second pulse is a red sideband 2π-pulse applied to ion B, coupling its ground state $|g\rangle_B$ to an auxiliary electronic state $|a\rangle_B$.

The relevant transition is

$$|g,1\rangle_B \longleftrightarrow |a,0\rangle_B .$$

A 2π-pulse completes a full Rabi oscillation, returning the population to its initial state with a phase factor of $(-i)^2 = -1$.

This occurs only when ion B is in $|g\rangle_B$ and a phonon is present. If ion B is in $|e\rangle_B$, the pulse does not couple to its electronic state.

Applying this pulse to the results of Question 2.1 gives

$$\boxed{\begin{aligned} |gg,0\rangle &\longrightarrow |gg,0\rangle\,, \\ |ge,0\rangle &\longrightarrow |ge,0\rangle\,, \\ -i\,|gg,1\rangle &\longrightarrow +i\,|gg,1\rangle\,, \\ -i\,|ge,1\rangle &\longrightarrow -i\,|ge,1\rangle\,. \end{aligned}}$$

Question 2.3

The third pulse is another red sideband π-pulse applied to ion A, reversing the population transfer from the first pulse:

$$|g,1\rangle_A \longrightarrow -i\,|e,0\rangle_A\,.$$

The combined motional state therefore returns to $|0\rangle_m$, leaving only the accumulated phases on the electronic states.

Applying this final pulse gives

$$\boxed{\begin{aligned} |gg,0\rangle &\longrightarrow |gg,0\rangle\,, \\ |ge,0\rangle &\longrightarrow |ge,0\rangle\,, \\ +i\,|gg,1\rangle &\longrightarrow |eg,0\rangle\,, \\ -i\,|ge,1\rangle &\longrightarrow -\,|ee,0\rangle\,. \end{aligned}}$$

Thus, the three-pulse sequence restores the collective motional ground state while introducing a phase of -1 only to the electronic state $|ee\rangle$.

Question 2.4

From the transformations in Question 2.3, the action on the electronic computational basis is

$$
\begin{aligned}
|gg\rangle &\longrightarrow |gg\rangle\,, \\
|ge\rangle &\longrightarrow |ge\rangle\,, \\
|eg\rangle &\longrightarrow |eg\rangle\,, \\
|ee\rangle &\longrightarrow -\,|ee\rangle\,.
\end{aligned}
$$

Therefore, in the ordered basis $\{|gg\rangle\,, |ge\rangle\,, |eg\rangle\,, |ee\rangle\}$, the resulting gate is

$$
\boxed{U = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} = CZ.}
$$

This is a controlled-Z gate, which applies a phase of -1 when both qubits are in their excited state.

It is an entangling gate because this conditional phase can transform an initially separable superposition into an entangled two-qubit state.

Question 2.5

A controlled-Z gate can be converted into a CNOT by applying Hadamard gates to ion B before and after the three-pulse entangling sequence:

$$
\boxed{\mathrm{CNOT}_{A\to B} = (I_A \otimes H_B)\, CZ\, (I_A \otimes H_B).}
$$

Physically, the Hadamard gates can be implemented using resonant carrier pulses on ion B, coupling $|g\rangle_B$ and $|e\rangle_B$ without changing the collective motional state. The pulse phases and durations are selected to produce the required single-qubit rotations.

Since

$$
HZH = X,
$$

the Hadamard gates convert the conditional phase flip of the CZ gate into a conditional bit flip. The resulting operation is a CNOT with ion A as the control and ion B as the target.

Question 2 – Part B(a)

The three-qubit extension of the controlled-Z gate is the controlled-controlled-Z (CCZ) gate. This applies a phase of -1 only when all three qubits are in their excited states $|e\rangle$, leaving all other computational basis states unchanged.

The required mapping is

$$\boxed{\begin{aligned}
|ggg\rangle &\longrightarrow |ggg\rangle \\
|gge\rangle &\longrightarrow |gge\rangle \\
|geg\rangle &\longrightarrow |geg\rangle \\
|gee\rangle &\longrightarrow |gee\rangle \\
|egg\rangle &\longrightarrow |egg\rangle \\
|ege\rangle &\longrightarrow |ege\rangle \\
|eeg\rangle &\longrightarrow |eeg\rangle \\
|eee\rangle &\longrightarrow -|eee\rangle
\end{aligned}}$$

Question 2 – Part B(b)

The CCZ gate can be implemented by extending the original three-pulse sequence to five pulses:

$$\boxed{\pi_A \longrightarrow \pi_B \longrightarrow 2\pi_C \longrightarrow \pi_B \longrightarrow \pi_A.}$$

The pulses on ion A couple $|g\rangle \leftrightarrow |e\rangle$, while those on ions B and C couple $|g\rangle \leftrightarrow |a\rangle$.

The first π-pulse on A creates a phonon if A is initially in $|e\rangle$. The subsequent π-pulse on B temporarily transfers this phonon into the auxiliary state $|a\rangle_B$ if B is in $|g\rangle$. Consequently, a phonon remains available only when both A and B were initially in $|e\rangle$.

The 2π-pulse on C introduces a phase of -1 if a phonon is present and C is in $|g\rangle$. The final π-pulses on B and A reverse the excitation transfers, restoring the collective motional ground state.

The accumulated phases cancel for all computational states except $|eee\rangle$, which acquires a phase of -1, producing the required CCZ gate.

The evolution of all eight initial states is shown in Table 1.

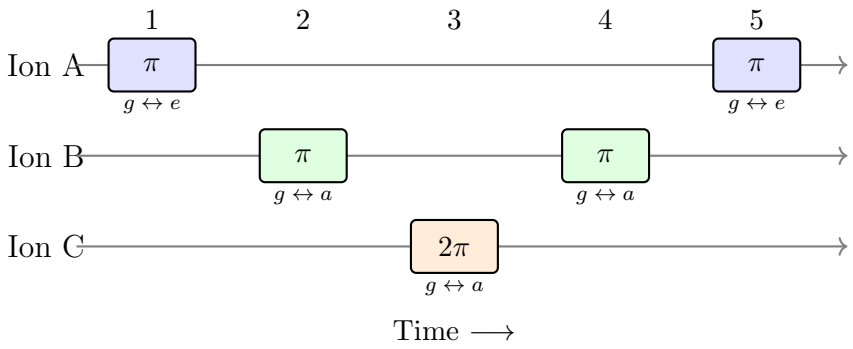


Figure 5: Five-pulse red sideband sequence implementing the CCZ gate using the shared collective motional mode.

Table 1: Evolution of the eight computational basis states through the five-pulse CCZ sequence.

Initial state	After π_A	After π_B	After $2\pi_C$	After π_B	After π_A
$\|ggg,0\rangle$	$\|ggg,0\rangle$	$\|ggg,0\rangle$	$\|ggg,0\rangle$	$\|ggg,0\rangle$	$\|ggg,0\rangle$
$\|gge,0\rangle$	$\|gge,0\rangle$	$\|gge,0\rangle$	$\|gge,0\rangle$	$\|gge,0\rangle$	$\|gge,0\rangle$
$\|geg,0\rangle$	$\|geg,0\rangle$	$\|geg,0\rangle$	$\|geg,0\rangle$	$\|geg,0\rangle$	$\|geg,0\rangle$
$\|gee,0\rangle$	$\|gee,0\rangle$	$\|gee,0\rangle$	$\|gee,0\rangle$	$\|gee,0\rangle$	$\|gee,0\rangle$
$\|egg,0\rangle$	$-i\,\|ggg,1\rangle$	$-\,\|gag,0\rangle$	$-\,\|gag,0\rangle$	$i\,\|ggg,1\rangle$	$\|egg,0\rangle$
$\|ege,0\rangle$	$-i\,\|gge,1\rangle$	$-\,\|gae,0\rangle$	$-\,\|gae,0\rangle$	$i\,\|gge,1\rangle$	$\|ege,0\rangle$
$\|eeg,0\rangle$	$-i\,\|geg,1\rangle$	$-i\,\|geg,1\rangle$	$i\,\|geg,1\rangle$	$i\,\|geg,1\rangle$	$\|eeg,0\rangle$
$\|eee,0\rangle$	$-i\,\|gee,1\rangle$	$-i\,\|gee,1\rangle$	$-i\,\|gee,1\rangle$	$-i\,\|gee,1\rangle$	$-\,\|eee,0\rangle$

As shown in the table, all computational states return to their original electronic configurations, and the collective motional mode returns to $|0\rangle$. Only the state $|eee\rangle$ acquires a negative phase:

$$\boxed{|eee, 0\rangle \longrightarrow -\,|eee, 0\rangle\,.}$$

The five-pulse sequence therefore implements the controlled-controlled-Z gate.

Question 3.1

The Lagrangian is given by $\mathcal{L} = T - V$, where the capacitors contribute kinetic energy and the Josephson junction contributes potential energy.

The node flux ϕ is defined such that $\dot{\phi} = V_A$. The capacitive energies are

$$T = \frac{1}{2}C_0\dot{\phi}^2 + \frac{1}{2}C_c(\dot{\phi} - V_g)^2.$$

The Josephson potential energy is

$$V = -E_J \cos\left(\frac{2\pi\phi}{\Phi_0}\right), \qquad \Phi_0 = \frac{h}{2e}.$$

Therefore, the Lagrangian is

$$\boxed{\mathcal{L} = \frac{1}{2}C_0\dot{\phi}^2 + \frac{1}{2}C_c(\dot{\phi} - V_g)^2 + E_J \cos\left(\frac{2\pi\phi}{\Phi_0}\right).}$$

Question 3.2

The momentum conjugate to the node flux ϕ is

$$q = \frac{\partial\mathcal{L}}{\partial\dot{\phi}}.$$

Differentiating the Lagrangian gives

$$\begin{aligned} q &= C_0\dot{\phi} + C_c(\dot{\phi} - V_g) \\ &= (C_0 + C_c)\dot{\phi} - C_cV_g. \end{aligned}$$

Let total capacitance $C_s = C_0 + C_c$,

$$\boxed{q = C_s\dot{\phi} - C_cV_g.}$$

The conjugate momentum represents the charge associated with the superconducting island.

Question 3.3

The Hamiltonian is obtained using the Legendre transformation

$$H = q\dot{\phi} - \mathcal{L}.$$

Expanding the Lagrangian from Question 3.1, and defining $C_s = C_0 + C_c$, gives

$$\mathcal{L} = \frac{1}{2}C_s\dot{\phi}^2 - C_cV_g\dot{\phi} + \frac{1}{2}C_cV_g^2 + E_J\cos\left(\frac{2\pi\phi}{\Phi_0}\right).$$

Substituting into the Legendre transformation,

$$\begin{aligned} H &= q\dot{\phi} - \frac{1}{2}C_s\dot{\phi}^2 + C_cV_g\dot{\phi} - \frac{1}{2}C_cV_g^2 - E_J\cos\left(\frac{2\pi\phi}{\Phi_0}\right) \\ &= \frac{1}{2}C_s\dot{\phi}^2 - \frac{1}{2}C_cV_g^2 - E_J\cos\left(\frac{2\pi\phi}{\Phi_0}\right), \end{aligned}$$

where we have used $q + C_cV_g = C_s\dot{\phi}$.
Substituting $\dot{\phi} = (q + C_cV_g)/C_s$ gives

$$H = \frac{(q + C_cV_g)^2}{2C_s} - E_J\cos\left(\frac{2\pi\phi}{\Phi_0}\right) - \frac{1}{2}C_cV_g^2.$$

Since V_g is fixed, the final term is a constant energy and can be omitted. Thus,

$$\boxed{H = \frac{(q + C_cV_g)^2}{2C_s} - E_J\cos\left(\frac{2\pi\phi}{\Phi_0}\right).}$$

To quantize the circuit, we promote ϕ and q to operators satisfying

$$[\hat{\phi}, \hat{q}] = i\hbar, \qquad \hat{q} = -i\hbar\frac{\partial}{\partial\phi}.$$

The time-independent Schrödinger equation is therefore

$$\boxed{\left[\frac{1}{2C_s}\left(-i\hbar\frac{\partial}{\partial\phi} + C_cV_g\right)^2 - E_J\cos\left(\frac{2\pi\phi}{\Phi_0}\right)\right]\psi(\phi) = E\psi(\phi).}$$

Question 3.4

For $V_g = 0$, the Hamiltonian from Question 3.3 becomes

$$H = \frac{q^2}{2C_s} - E_J \cos\left(\frac{2\pi\phi}{\Phi_0}\right),$$

where $C_s = C_0 + C_c$.

Near the potential minimum at $\phi = 0$, use the harmonic approximation

$$\cos\left(\frac{2\pi\phi}{\Phi_0}\right) \approx 1 - \frac{1}{2}\left(\frac{2\pi\phi}{\Phi_0}\right)^2.$$

The Hamiltonian therefore becomes

$$H \approx \frac{q^2}{2C_s} + \frac{E_J}{2}\left(\frac{2\pi}{\Phi_0}\right)^2 \phi^2 - E_J.$$

Comparing with the harmonic oscillator Hamiltonian

$$H_{\mathrm{HO}} = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2,$$

we identify

$$m = C_s, \qquad m\omega^2 = E_J\left(\frac{2\pi}{\Phi_0}\right)^2.$$

Hence, the oscillator frequency is

$$\boxed{\omega = \frac{2\pi}{\Phi_0}\sqrt{\frac{E_J}{C_s}}.}$$

The harmonic oscillator energy levels, including the constant potential offset $-E_J$, are

$$E_n \approx -E_J + \hbar\omega\left(n + \frac{1}{2}\right).$$

Therefore, the ground and first excited state energies are

$$\boxed{\begin{aligned} E_0 &\approx -E_J + \frac{\hbar\omega}{2}, \\ E_1 &\approx -E_J + \frac{3\hbar\omega}{2}. \end{aligned}}$$

Using $\Phi_0 = h/(2e)$, the oscillator energy becomes

$$\hbar\omega = \hbar\frac{2\pi}{\Phi_0}\sqrt{\frac{E_J}{C_s}} = 2e\sqrt{\frac{E_J}{C_s}}.$$

Thus, the ground and first excited state energies are

$$\boxed{\begin{aligned} E_0 &\approx -E_J + e\sqrt{\frac{E_J}{C_s}}, \\ E_1 &\approx -E_J + 3e\sqrt{\frac{E_J}{C_s}}. \end{aligned}}$$

The corresponding energy gap is

$$\boxed{E_1 - E_0 \approx 2e\sqrt{\frac{E_J}{C_s}}.}$$

Question 3.5

From Question 3.3, the Hamiltonian is

$$H = \frac{(\hat{q} + C_c V_g)^2}{2C_s} - E_J \cos\left(\frac{2\pi\hat{\phi}}{\Phi_0}\right), \qquad C_s = C_0 + C_c.$$

To calculate the energy levels numerically, express the Hamiltonian in the Cooper-pair number basis $\{|n\rangle\}$, where n is the number of excess Cooper pairs on the superconducting island.

Since each Cooper pair carries charge $2e$,

$$\hat{q}\,|n\rangle = 2en\,|n\rangle\,.$$

The charging-energy term is therefore diagonal. For the Josephson term, using Euler's identity,

$$-E_J \cos\left(\frac{2\pi\hat{\phi}}{\Phi_0}\right) = -\frac{E_J}{2}\left(e^{i2\pi\hat{\phi}/\Phi_0} + e^{-i2\pi\hat{\phi}/\Phi_0}\right).$$

The exponential operators increase or decrease the Cooper-pair number by one, corresponding to tunnelling through the Josephson junction:

$$e^{\pm i2\pi\hat{\phi}/\Phi_0}\,|n\rangle = |n \pm 1\rangle\,.$$

The Hamiltonian acting on a number state is therefore

$$\boxed{H\,|n\rangle = \frac{(2en + C_c V_g)^2}{2C_s}\,|n\rangle - \frac{E_J}{2}\left(|n+1\rangle + |n-1\rangle\right).}$$

Thus, in the number basis, the Hamiltonian has diagonal charging-energy terms and off-diagonal Josephson couplings of $-E_J/2$ between neighbouring states.

Truncate the basis to $n = -10, \ldots, 10$, producing a 21×21 tridiagonal matrix. Using the specified ratio $E_{C0}/E_J = 1$, numerically calculate the four lowest eigenvalues as functions of V_g using Mathematica.

```
1
2
3  (* Parameters: EC0/EJ = 1 *)
4  EJ = 1;
5  e = 1;
6  C0 = e^2/(2 EJ);
7  Cc = C0/10;
8  Cs = C0 + Cc;
9
10 (* Hamiltonian in the basis n = -10,...,10 *)
11 H[Vg_] := DiagonalMatrix[
12     Table[(2 e n + Cc Vg)^2/(2 Cs), {n, -10, 10}]
13     ] - EJ/2 (
14     DiagonalMatrix[ConstantArray[1, 20], 1] +
15     DiagonalMatrix[ConstantArray[1, 20], -1]
16     );
17
18 (* Four lowest eigenvalues *)
19 energies[Vg_?NumericQ] :=
20   Take[Sort[Eigenvalues[N[H[Vg]]]], 4];
```

```
21
22 (* Plot energy levels over one gate-voltage period *)
23 data = Table[
24    Join[{v}, energies[v]],
25    {v, 0, 2 e/Cc, (2 e/Cc)/100}
26    ];
27
28 ListLinePlot[
29  Table[data[[All, {1, k}]], {k, 2, 5}],
30  AxesLabel -> {"Vg", "Energy / EJ"},
31  PlotLegends -> {"E0", "E1", "E2", "E3"},
32  PlotRange -> All
33  ]
34
35 gap[Vg_?NumericQ] := energies[Vg][[2]] - energies[Vg][[1]];
36
37 NMinimize[{gap[Vg], 0 <= Vg <= 2 e/Cc}, Vg]
```

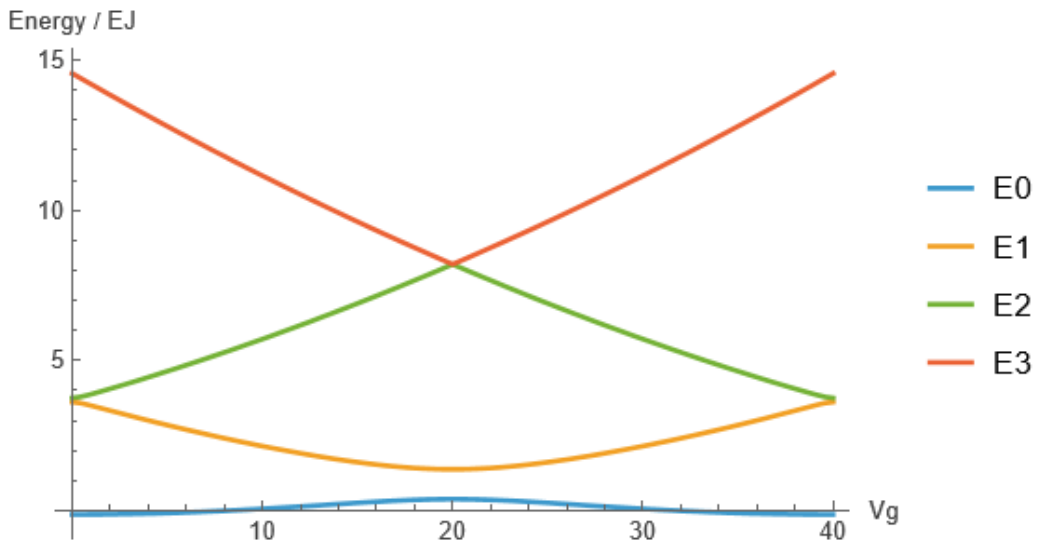


Figure 6: The four lowest energy eigenvalues of the superconducting circuit as functions of the applied gate voltage V_g, calculated by diagonalising the Hamiltonian in the truncated Cooper-pair number basis.

Question 3.6

The energy gap between the ground and first excited states is

$$\Delta E(V_g) = E_1(V_g) - E_0(V_g).$$

Using Mathematica's `NMinimize` function with the eigenvalues calculated in Question 3.5,

$$\boxed{\Delta E_{\min} = 0.995301\, E_J, \qquad V_g = 20.}$$

Question 3 – Bonus

The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel Devoret and John Martinis for demonstrating macroscopic quantum mechanical tunnelling and energy quantisation in superconducting electrical circuits. Their experiments showed that collective electrical variables, such as the phase difference across a Josephson junction, can exhibit quantum behaviour. This is directly related to the Cooper-pair box studied in this question, where the charge and flux are treated as conjugate quantum variables. The resulting Hamiltonian produces discrete energy levels, demonstrating how a superconducting circuit can behave as an artificial atom and form the basis of a superconducting qubit.

Question 4(a)

Label the input ports of the first beam splitter a, b, the two interferometer arms u, l (upper and lower), and the final output ports c, d, leading to detectors D_1, D_2 respectively.

1. Input state The input is the antisymmetric polarisation Bell state

$$\boxed{|\Psi^-\rangle_{ab} = \frac{1}{\sqrt{2}}\left(a_H^\dagger b_V^\dagger - a_V^\dagger b_H^\dagger\right)|0\rangle\,,}$$

where $a_H^\dagger$, for example, creates a horizontally polarised photon in spatial mode a.

2. After the first beam splitter For a 50:50 beam splitter, we choose the transformations

$$a_\lambda^\dagger \longrightarrow \frac{u_\lambda^\dagger + l_\lambda^\dagger}{\sqrt{2}},$$
$$b_\lambda^\dagger \longrightarrow \frac{u_\lambda^\dagger - l_\lambda^\dagger}{\sqrt{2}},$$

where $\lambda = H, V$.

Applying these to the input state gives

$$|\Psi^-\rangle_{ab} \longrightarrow \frac{1}{2\sqrt{2}}\Big[(u_H^\dagger + l_H^\dagger)(u_V^\dagger - l_V^\dagger) - (u_V^\dagger + l_V^\dagger)(u_H^\dagger - l_H^\dagger)\Big]|0\rangle\,.$$

Expanding and collecting terms,

$$= \frac{1}{2\sqrt{2}}\Big[u_H^\dagger u_V^\dagger - u_H^\dagger l_V^\dagger + l_H^\dagger u_V^\dagger - l_H^\dagger l_V^\dagger - u_V^\dagger u_H^\dagger + u_V^\dagger l_H^\dagger - l_V^\dagger u_H^\dagger + l_V^\dagger l_H^\dagger\Big]|0\rangle\,.$$

Since photon creation operators commute, the terms corresponding to two photons in the same arm cancel. The remaining terms give

$$\boxed{|\Psi^-\rangle_{ul} = \frac{1}{\sqrt{2}}\left(u_H^\dagger l_V^\dagger - u_V^\dagger l_H^\dagger\right)|0\rangle\,,}$$

up to an overall phase.

Thus, after BS_1, there is exactly one photon in each interferometer arm.

3. After the second beam splitter The mirrors direct the two arms into BS_2. Assuming polarisation-independent phase shifts, the Bell state is unchanged apart from an overall phase.

At BS_2, we apply the same 50:50 transformations:

$$u_\lambda^\dagger \longrightarrow \frac{c_\lambda^\dagger + d_\lambda^\dagger}{\sqrt{2}},$$
$$l_\lambda^\dagger \longrightarrow \frac{c_\lambda^\dagger - d_\lambda^\dagger}{\sqrt{2}}.$$

Substituting into the state after BS_1,

$$|\Psi^-\rangle_{ul} \longrightarrow \frac{1}{2\sqrt{2}}\Big[(c_H^\dagger + d_H^\dagger)(c_V^\dagger - d_V^\dagger) - (c_V^\dagger + d_V^\dagger)(c_H^\dagger - d_H^\dagger)\Big]|0\rangle\,.$$

As before, the terms with both photons in the same output port cancel. The final state is therefore

$$\boxed{|\Psi^-\rangle_{cd} = \frac{1}{\sqrt{2}}\left(c_H^\dagger d_V^\dagger - c_V^\dagger d_H^\dagger\right)|0\rangle\,,}$$

again up to an overall phase.

4. Detection pattern The final state contains one photon in each output mode, with either H or V polarisation.

Consequently, the detectors always register a coincidence:

$$\boxed{P(D_1 = 1, D_2 = 1) = 1.}$$

The probabilities of detecting both photons at the same detector are

$$\boxed{P(D_1 = 2, D_2 = 0) = P(D_1 = 0, D_2 = 2) = 0.}$$

Thus, the antisymmetric Bell state exhibits perfect antibunching at both beam splitters, resulting in one photon at each detector.

Question 4(b)

We use the same 50:50 beam-splitter transformations as in Question 4(a):

$$a_\lambda^\dagger \longrightarrow \frac{c_\lambda^\dagger + d_\lambda^\dagger}{\sqrt{2}},$$
$$b_\lambda^\dagger \longrightarrow \frac{c_\lambda^\dagger - d_\lambda^\dagger}{\sqrt{2}},$$

where a, b are the input modes, c, d are the output modes, and $\lambda = H, V$. For two identical photons in the same mode, we use

$$(c_\lambda^\dagger)^2 \left|0\right\rangle = \sqrt{2}\left|2\lambda, 0\right\rangle, \qquad (d_\lambda^\dagger)^2 \left|0\right\rangle = \sqrt{2}\left|0, 2\lambda\right\rangle.$$

1. The state $|\Psi^+\rangle$ The input state is

$$\left|\Psi^+\right\rangle_{ab} = \frac{1}{\sqrt{2}}\left(a_H^\dagger b_V^\dagger + a_V^\dagger b_H^\dagger\right)\left|0\right\rangle.$$

Applying the beam-splitter transformations,

$$\left|\Psi^+\right\rangle_{ab} \longrightarrow \frac{1}{2\sqrt{2}}\Big[(c_H^\dagger + d_H^\dagger)(c_V^\dagger - d_V^\dagger) + (c_V^\dagger + d_V^\dagger)(c_H^\dagger - d_H^\dagger)\Big]\left|0\right\rangle.$$

Expanding and using the commutation of photon creation operators, the terms involving photons in different output ports cancel, leaving

$$\frac{1}{\sqrt{2}}\left(c_H^\dagger c_V^\dagger - d_H^\dagger d_V^\dagger\right)\left|0\right\rangle.$$

Therefore,

$$\boxed{\left|\Psi^+\right\rangle_{\text{out}} = \frac{1}{\sqrt{2}}\left(\left|HV, 0\right\rangle - \left|0, HV\right\rangle\right).}$$

Both photons leave through the same output port, with orthogonal polarisations.

2. The state $|\Phi^+\rangle$ The input state is

$$|\Phi^+\rangle_{ab} = \frac{1}{\sqrt{2}}\left(a_H^\dagger b_H^\dagger + a_V^\dagger b_V^\dagger\right)|0\rangle\,.$$

Applying the beam-splitter transformations,

$$|\Phi^+\rangle_{ab} \longrightarrow \frac{1}{2\sqrt{2}}\Big[(c_H^\dagger + d_H^\dagger)(c_H^\dagger - d_H^\dagger) + (c_V^\dagger + d_V^\dagger)(c_V^\dagger - d_V^\dagger)\Big]|0\rangle\,.$$

Since creation operators commute, the cross terms cancel, giving

$$\frac{1}{2\sqrt{2}}\left[(c_H^\dagger)^2 - (d_H^\dagger)^2 + (c_V^\dagger)^2 - (d_V^\dagger)^2\right]|0\rangle\,.$$

Using the normalisation of two-photon Fock states,

$$\boxed{|\Phi^+\rangle_{\text{out}} = \frac{1}{2}\left(|2H,0\rangle - |0,2H\rangle + |2V,0\rangle - |0,2V\rangle\right).}$$

Thus, both photons leave through the same port and have identical polarisations.

3. The state $|\Phi^-\rangle$ The input state is

$$|\Phi^-\rangle_{ab} = \frac{1}{\sqrt{2}}\left(a_H^\dagger b_H^\dagger - a_V^\dagger b_V^\dagger\right)|0\rangle\,.$$

The calculation is identical to that for $|\Phi^+\rangle$, except for the relative minus sign between the polarisation terms:

$$|\Phi^-\rangle_{ab} \longrightarrow \frac{1}{2\sqrt{2}}\Big[(c_H^\dagger + d_H^\dagger)(c_H^\dagger - d_H^\dagger) - (c_V^\dagger + d_V^\dagger)(c_V^\dagger - d_V^\dagger)\Big]|0\rangle\,.$$

Expanding and cancelling the cross terms,

$$\frac{1}{2\sqrt{2}}\left[(c_H^\dagger)^2 - (d_H^\dagger)^2 - (c_V^\dagger)^2 + (d_V^\dagger)^2\right]|0\rangle\,.$$

Therefore,

$$\boxed{|\Phi^-\rangle_{\text{out}} = \frac{1}{2}\left(|2H,0\rangle - |0,2H\rangle - |2V,0\rangle + |0,2V\rangle\right).}$$

Again, both photons leave through the same output port, with identical polarisations.

Question 4 – Part B

The detection pattern is a consequence of two-photon quantum interference closely related to the Hong–Ou–Mandel (HOM) effect, although it differs from the conventional HOM result.

In the standard HOM experiment, two indistinguishable photons entering separate ports of a 50:50 beam splitter interfere destructively in the coincidence outcomes. Consequently, the photons bunch and leave through the same output port.

In this experiment, however, the input is the antisymmetric Bell state

$$|\Psi^-\rangle_{12} = \frac{1}{\sqrt{2}}\left(|1H,2V\rangle - |1V,2H\rangle\right).$$

The relative minus sign causes destructive interference between the amplitudes for both photons leaving through the same port. Consequently, the photons antibunch, producing one photon in each output mode at both beam splitters.

The observed antibunching is a HOM-type two-photon interference effect arising from the antisymmetric polarisation state, rather than the conventional HOM bunching effect.

However, the final coincidence pattern alone does not demonstrate HOM interference. Experimentally, one could vary the relative arrival time of the photons to test whether the coincidence probability depends on their temporal indistinguishability.